

ZS-A24

Dimension-Weighted Mediator Semigroups and Their Spin-Graded Continuous-Core Lift

Finite-register modular unification, with the continuous-core dynamical lift left open (F-A24.9).

Part I — the general mediator-semigroup theorem, the exact H-theorem, the full 121×121 Liouvillian spectrum $\{0, -2A/Q, -A/2, -9A/Q, -A\}$, and perturbation stability. Part II — the genuine Type III $\rightarrow$ Type II $_{\infty}$ continuous core $C_{\omega} = \mathcal{A}^{III} \rtimes_{\sigma} \mathbb{R}$, its finite Type II $_1$ corner $eC_{\omega}e$, and the finite-register embedding $\iota : M_{11} \hookrightarrow eC_{\omega}e$; the de Sitter observer trace and the CLPW offset (embedding-conditional); the face-embedding frontier redefined by trace dimension; and a two-edge instanton whose factor 2 is DERIVED and whose per-edge normalisation is reduced to a sharp condition. An honest conditional lift that retracts the v1.0 finite-route Type II claim.

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Repository: github.com/KennyKang-git/zspin

Verification: 54/54 computational, regression, and consistency checks PASS (zs_a24_v2_1_audit.py, attached and runnable) | Zero Free Parameters. All quantities derive from $A = 35/437$, $Q = 11$, and $(d_Z, d_X, d_Y) = (2, 3, 6)$; $\kappa^2 = A/Q$. The checks are symbolic, regression, and consistency verifications (not 53 separate theorem proofs); the theorem-level results carry their own proofs in the text and appendices, and the per-version audit trail is in Appendix E and the Version History.

Sole geometric inputs: $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$, **LOCKED**, with $\kappa^2 = A/Q = 35/4807$ and $1/\kappa^2 = Q/A = 4807/35$ (PROVEN, ZS-A19 / ZS-M6 §2.2). Units: reduced Planck mass $\bar{M}_P^2 = (8\pi G)^{-1}$; $\bar{H}_0/\bar{M}_P \approx 5.9 \times 10^{-61}$ is a disclosed external normalisation, **not** derived (= the corpus-wide B3 debt). A24 *consumes* the locked anchors and introduces **zero** new fitted parameters. $(A, Q, \dim Z) = (35/437, 11, 2)$ **LOCKED**.

This is v2.0 — the consolidated public release. It presents the final state of the theory in three Parts and is the result of the audit trail in Appendix E and the Version History (v1.0 $\rightarrow$ v2.0); the per-version corrections are not relitigated in the main body. The three results:

(I) Part I — an exact open-system theory. The dimension-weighted mediator semigroup has a general connected-graph stationary law, an exact H-theorem, a unique sector-covariant GKLS realisation, an **exact general Liouvillian spectrum** (Theorem A24.I-4G, any (a, b, c) ; Z-Spin $\{0, -2A/Q, -A/2, -9A/Q, -A\}$, multiplicities $(1, 80, 36, 3, 1)$), and **two PROVEN perturbation theorems**. Every rate is a rational multiple of A ; zero new parameters.

(II) Part II — the honest continuous-core lift. The finite Type-II route is **RETRACTED** (inner modular flow); the genuine core is the Type III $_1 \rightarrow$ II $_{\infty}$ crossed product $C_{\omega} = \mathcal{A}^{III} \rtimes_{\sigma} \mathbb{R}$ (IMPORTED-PROVEN), with a II $_1$ corner and a register embedding $\iota : M_{11} \hookrightarrow M_{\text{obs}} \subset C_{\omega}$ (DERIVED-CONDITIONAL). The corner *trace* is the matrix trace $= \pi = (3, 2, 6)/11$; the observer weight $\omega = (9, 4, 36)/49$ is the *state* $\tau(h \cdot)$.

(III) Part III — the modular half-density bridge, at the finite-register level. L_π (stationary $\pi \propto d_i$) and L_ω (stationary $\omega \propto d_i^2$) are two reversible dynamics on M_{11} joined by a modular half-density transform — the $s = 0$ and $s = 1/2$ endpoints of one **microscopic QMS interpolation family** $L_{\{(s)\}}$ (Theorem A24.III-2), with L_ω proven KMS- and GNS-symmetric and modular-covariant (Theorem A24.III-1b). A no-go forbids a single generator from balancing both. The **continuous-core dynamical lift** — a generator $\mathcal{L}_s$ on M_{obs} with $E \circ \mathcal{L}_s = L_{s \circ E}$, modular covariance and complete positivity — is the **single honest OPEN item (F-A24.9)**, consistent with the Book's open Condition C.

Honest scope: the unification is **finite-register**; the standard tilted-Lindblad / KMS–GNS / Dirichlet machinery is credited as standard, the novelty being the exact package on the 3-2-6 register. The runnable audit reports **54/54 computational, regression, and consistency checks PASS**.

§0. Abstract

ZS-A23 reduced the dynamical heart of the corpus to one dimension-weighted mediator generator $q_{\{i \rightarrow j\}} = \Gamma_0 \kappa^2 d_j$, built from the single coupling $\kappa^2 = A/Q$. This paper is its three-Part development: a self-contained general theory of such semigroups (Part I), an honest lift of the finite register to a genuine de Sitter observer algebra (Part II), and the finite-register modular half-density bridge joining the state weight π to the operator weight ω (Part III).

Part I (PROVEN). For *any* finite connected graph with node multiplicities d_i , the generator $q_{\{i \rightarrow j\}} = \Gamma_0 \kappa^2 a_{\{ij\}} d_j$ has stationary distribution $\pi_i = d_i / \sum_j d_j$, is reversible, obeys modular detailed balance, and has an exact H-theorem (Theorems A24.I-1–3). The sector-covariant GKLS realisation is unique (Schur per edge, plus global edge-homogeneity identifying the two edge scalars with the single κ^2). Its **full 121×121 Liouvillian** closes with no new parameters:

$$\text{Spec}(\mathcal{L}) = \{0, -2A/Q, -A/2, -9A/Q, -A\} \text{ with multiplicities } (1, 80, 36, 3, 1), \Sigma = 121,$$

the cross-sector coherences decaying at the clean rate $A/2$ (Theorem A24.I-4G — an exact spectrum for any sector dimensions). The generator is perturbation-stable with two PROVEN theorems (a reciprocal Dirichlet-form gap bound and an exact directional 3-state analysis), giving explicit simulator tolerance curves.

Part II (the honest lift). v1.0 claimed $M_{11} \rtimes_{\{\sigma^\wedge \omega\}} \mathbb{R}$ is Type II. This is **false**: M_{11} is a finite factor, its modular flow $\sigma_t^\wedge \omega = \text{Ad}(\rho_\omega^{\wedge \{it\}})$ is *inner* (Skolem–Noether), and the crossed product by an inner action is $M_{11} \rtimes^{\text{inner}} L^\infty(\mathbb{R})$ — Type I with diffuse centre, not a II factor (Theorem A24.II-0, **CLOSED-NEGATIVE for the finite route**). The genuine construction (Witten/CLPW/Takesaki) starts from the **Type III₁ QFT algebra** $\mathcal{A}^{\text{III}}$, whose modular flow is outer; $C_\omega := \mathcal{A}^{\text{III}} \rtimes_{\sigma_\omega} \mathbb{R}$ is **Type II_∞** (Theorem A24.II-1, IMPORTED-PROVEN). A finite-trace corner $M_{\text{obs}} := e C_\omega e$ ($0 < \tau(e) < \infty$) is **Type II₁**, and the finite register embeds, $\iota : M_{11} \hookrightarrow M_{\text{obs}}$ (Theorem A24.II-2, DERIVED-CONDITIONAL). The corner trace is the **matrix trace**: the **11 Hilbert microstate projections** e_α each carry $\tau(e_\alpha) = 1/D = 1/11$, so $\tau(P_i) = d_i/D = (3, 2, 6)/11 = \pi$; the **121 operator-slot projections** q_a (the face-embedding targets) carry $\tau(q_a) = 1/121$. The operator weight $\omega = (9, 4, 36)/49$ is the **state** $\omega(\cdot) = \tau(h \cdot)$ defined by the modular density h (PART III), and the CLPW offset $A_{\text{dS}}/4G$ enters Gate 2(ii) (DERIVED-CONDITIONAL / IMPORTED-MATCHING). The

face-embedding target is realised in the Π_1 corner: $\Phi_{\text{face}} : \mathbb{C}^{121} \rightarrow M_{\text{obs}}$ sends 121 minimal idempotents to orthogonal projections of *continuous trace* $\tau(q_a) = 1/121$ (not rank in M_{11} , which is impossible), and **v1.2 grounds the partition in ZS-F2 §11.4** — the coarse traces (6, 32, 83)/121 are (F(cube), F(truncated icosahedron), remainder), so **Gate 3 advances OPEN** →

DERIVED-CONDITIONAL; **v1.3 resolves the equal-weight question** in favour of **combinatorial** (not metric) holography, since ZS-F0 §8.5 makes the Z-sector partition function a BV–BFV cobordism invariant (topological). The Gate-2 observer weight (9, 4, 36)/49 is **PROVEN** as the operator-sampling weight $d_i^2/\Sigma d_j^2$ (ZS-A23.MC), but **v1.5 corrects its operator-algebra identity**: the Π_1 -corner **trace** is the matrix trace $= d_i/D = (3, 2, 6)/11 = \pi$ (the Part-I stationary), while $\omega = (9, 4, 36)/49$ is the **state** $\tau(h \cdot)$ given by the modular density $h_i = D d_i/\Sigma d_j^2$ (v1.4 had conflated this state with the trace, contradicting its own audit and ZS-A23). Read correctly, the Takesaki dual weight is the trace times h , so ω is the dual-weight *state*. This delivers the **complete finite-register trace/state unification (PART III)**: the dimension-weighted L_π (stationary π) and the observer-balanced L_ω (stationary ω) are two reversible dynamics on the *same register algebra* M_{11} , joined by a **modular half-density transform** and embedded in one interpolation family $L_{\{s\}}$, $s \in [0, 1/2]$ (Central Theorem A24.III; a no-go forbids a single generator from balancing both, and **v1.6 proves** L_ω is both KMS- and GNS-symmetric on M_{11} with $[L_\omega, \sigma_t^\omega] = 0$). This supplies the **candidate finite-dimensional dynamics to be lifted to the continuous core**; the lift itself — a generator $\mathcal{L}_s$ on $M_{\text{obs}} = eC_\omega e$ intertwining the conditional expectation with modular covariance and complete positivity — is **OPEN (F-A24.9)**, consistent with the Book/ZS-F23 keeping Condition C open. The two-edge instanton **factor 2 is DERIVED**; **v1.3 corrects the per-edge value**: $1/\kappa^2$ is **RETRACTED** (no corpus instanton equals it — the corpus actions are π/A , $2\pi/A$, $5\pi/A$, $35\pi/3$), and since the A24 mediation crosses the $\mathbb{Z}_2$ seam (ZS-F23), the per-edge action is the **DERIVED seam-flip $t_{\text{flip}} = \pi/A$** (ZS-M3 §6), giving $S_{\{X \rightarrow Y\}} = 2\pi/A = T_{\text{micro}}$. The old $e^{\{-2/\kappa^2\}} \approx \Lambda$ motivation is exposed as numerology and **RETRACTED** (Gate 4, §14).

Net. The semigroup is now a complete open-quantum-systems theory (Part I); the continuous-core lift is on a correct algebraic footing (Part II); the cosmological-constant problem remains localised to the embedding, the B3 scale, and the per-edge instanton normalisation, with every over-claim of v1.0 retracted or made conditional. Zero free parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ **LOCKED**.

Epistemic Status Legend

Table 1. Epistemic status legend.

STATUS	DEFINITION
PROVEN	Mathematical theorem; standard mathematics alone, machine-verifiable.
PROVEN-with-hypotheses	Theorem under explicitly stated physical hypotheses (named in situ).
IMPORTED-PROVEN	Result proved externally, used without re-proof; full citation given.
IMPORTED-MATCHING	An external result invoked with a chosen reference-state normalisation; a matching, not an independent derivation.
DERIVED	Z-Spin action plus standard physics; zero free parameters.
DERIVED-CONDITIONAL	DERIVED conditional on listed hypotheses not themselves closed.
PREDICTION	A falsifiable consequence stated as such.

STATUS	DEFINITION
HYPOTHESIS-strong / weak	Motivated conjecture; documented closure route / failing chain.
OBSERVATION / CONSISTENCY	A numerical agreement reported as such, not a derivation.
NON-CLAIM	Explicit declaration of what is NOT asserted.
OPEN	Recognised gap; an executable closure route is named where one exists.
CLOSED-NEGATIVE	A registered gate / sub-route resolved in the negative (a theorem).
RETRACTED	A prior-version claim withdrawn as unsupported, with reason.
LOCKED	Core constant fixed upstream; immutable downstream.

§1. Introduction

ZS-A23 established an exact correspondence between the Z-Spin-mediated finite-register operator algebra and a reversible three-sector Markov semigroup, and named (A23 §13) its own continuation: "the unified spin-graded continuous-clock program (ZS-A24)." Version 1.0 of this paper attempted that program but made a foundational error, identified in external review and corrected here.

The v1.0 error and its correction. v1.0 asserted (Theorem A24.1a) that the crossed product $\mathbf{M}_C = M_{11} \rtimes_{\{\sigma^\omega\}} \mathbb{R}$ is a Type II von Neumann algebra. This is incorrect. M_{11} is a finite-dimensional factor; by the Skolem–Noether theorem every automorphism of M_{11} is inner, and the modular automorphism v1.0 itself wrote, $\sigma_t^\omega(x) = \rho_\omega\{it\} x \rho_\omega\{-it\} = \text{Ad}(\rho_\omega\{it\})(x)$, is manifestly inner. The crossed product of a von Neumann algebra by an *inner* one-parameter action is the *tensor product* with the group algebra, $M_{11} \rtimes_{\{\text{Ad } \rho^\omega\}} \mathbb{R} \cong M_{11} \otimes L^\infty(\mathbb{R})$, a Type I algebra with diffuse centre $1 \otimes L^\infty(\mathbb{R})$ — **not a Type II factor**. v1.0's falsification gate F-A24.1 stated exactly this kill-switch ("if σ^ω is inner, Type I results and Theorem A24.1(a) fails") and then mis-cleared it by confusing *non-trivial* (the flow is non-trivial because the d_i differ) with *outer* (a non-trivial inner action is still inner). **F-A24.1 is therefore triggered, and the finite-route Type II claim is RETRACTED** (§9, CLOSED-NEGATIVE).

The genuine construction — Witten's "gravity and the crossed product" and CLPW's de Sitter observer algebra — starts from the **Type III₁** algebra of quantum-field observables, whose modular flow *is* outer, and produces a **Type II_∞** factor by Takesaki's continuous decomposition. The finite register M_{11} is not the source of the Type II structure; it is *embedded* inside the genuine continuous core. v1.1 rebuilds Part II on this correct footing: the continuous core $\mathbf{C}_\omega = \mathcal{A}_{\text{III}} \rtimes_\sigma \mathbb{R}$ (II_∞) contains a finite corner $e\mathbf{C}_\omega e$ (II₁), into which the register embeds, $\iota : M_{11} \hookrightarrow e\mathbf{C}_\omega e \subset \mathbf{C}_\omega$.

The three-Part structure. A23 named the dimension-weighted mediator semigroup — algebra generation, uniqueness, modular detailed balance, the H-theorem, the dimension-weighted Laplacian, the simulator — as its strongest stand-alone result. This paper develops it as a self-contained **Part I** (with the **full 121×121 Liouvillian spectrum** — the coherence modes, not only the 3×3 population sector — and **perturbation stability**, the tolerance curves a real simulator needs); **Part II** is the spin-graded continuous-core lift; and **Part III** is the finite-register modular half-density bridge joining the state weight

π to the operator weight ω . This keeps the one-paper policy while ensuring no programme is buried in another.

Scope and posture. This is v2.0, the consolidated public release. Part I is PROVEN, with an exact general spectrum (Theorem A24.I-4G) and two PROVEN perturbation theorems. In Part II the II_∞ core is IMPORTED-PROVEN; the Gate-2 trace/state identity is correct (corner *trace* = matrix trace = $\pi = (3, 2, 6)/11$; observer weight $\omega = (9, 4, 36)/49$ = the *state* $\tau(h \cdot)$). PART III proves the **finite-register** unification — L_π and L_ω are the $s = 0, \frac{1}{2}$ endpoints of one **microscopic QMS interpolation family** $\text{L}_{\{(s)\}}$ on M_{11} (Theorem A24.III-2), with L_ω KMS- and GNS-symmetric and modular-covariant (Theorem A24.III-1b). The honest scope is *finite-register* unification plus a *candidate dynamics to be lifted*; the **continuous-core dynamical lift is OPEN** (F-A24.9), the single honest gap. The surviving residuals are explicitly **upstream** (ZS-F0's deferred state-sum, CLPW's reference state), **corpus-wide** (B3), or **registered** (C5). No new free parameter is introduced, and the standard tilted-Lindblad / KMS–GNS / Dirichlet machinery is credited as standard (the novelty is the exact package on the 3-2-6 register).

§2. Locked Inputs and Conventions

Time-normalisation convention. Throughout, time is measured in units of the common microscopic attempt rate Γ_0 : $t_{\text{dimensionless}} = \Gamma_0 t$, so that $\Gamma_0 \equiv \mathbf{1}$ in all quoted spectra and simulator rates (the dimension-weighted rate is $q_{\{i \rightarrow j\}} = \Gamma_0 \kappa^2 d_j$, and with $\Gamma_0 = 1$ it is $\kappa^2 d_j$). Γ_0 **is therefore a unit convention, not a fitted parameter** — every reported eigenvalue carries an implicit overall factor Γ_0 , e.g. the population spectrum is $\Gamma_0 \cdot \{0, -2\mathbf{A}/\mathbf{Q}, -\mathbf{A}/2, -9\mathbf{A}/\mathbf{Q}, -\mathbf{A}\}$. This is what "zero free parameters" means here: the *ratios* and the *dimensionless* structure are fixed entirely by $\mathbf{A}$, $\mathbf{Q}$, and the sector dimensions; Γ_0 only sets the clock.

$(\mathbf{A}, \mathbf{Q}, (Z, X, Y)) = (35/437, 11, (2, 3, 6))$ **LOCKED**. $\kappa^2 = \mathbf{A}/\mathbf{Q} = 35/4807$; $1/\kappa^2 = \mathbf{Q}/\mathbf{A} = 4807/35$ (PROVEN, ZS-A19; ZS-M6 §2.2). Units: $\bar{M}_P^2 = (8\pi G)^{-1}$; $\bar{H}_0/\bar{M}_P \approx 5.9 \times 10^{-61}$ (NON-CLAIM, = B3). Sector dimensions $(d_X, d_Z, d_Y) = (a, b, c) = (3, 2, 6)$, $D = a+b+c = \mathbf{Q} = 11$.

Inherited from A23 (PROVEN): finite dynamical algebra $\mathcal{A}_{\text{phys}} = M_{11}(\mathbb{C})$ (Hilbert dim 11, operator dim 121); observer algebra $\mathbf{N} = M_3 \oplus M_2 \oplus M_6$ (operator dim 49); the mediator graph is the path $X-Z-Y$ with $\text{L}_{XY} \equiv 0$; stationary $\pi = (3, 2, 6)/11$; population spectrum $\{0, -2\mathbf{A}/\mathbf{Q}, -\mathbf{A}\}$; observer weights $\omega = (9, 4, 36)/49$; RN density $h_i = 11d_i/49$; modular flow $\sigma_{t^*} \omega(C_{\{ij\}}) = (d_i/d_j)^{\{it\}} C_{\{ij\}}$; modular centralizer = $\mathbf{N}$; $e^{\{-c\}} = 121/49 = \dim M_{11}/\dim \mathbf{N}$; index 3.

New objects (Part II): the Type III_1 field algebra $\mathcal{A}^{\text{III}}$; its outer modular flow σ_t ; the continuous core $\text{C}_\omega = \mathcal{A}^{\text{III}} \rtimes_\sigma \mathbb{R}$ (II_∞) with trace τ ; a finite projection e ($0 < \tau(e) < \infty$); the finite corner $M_{\text{obs}} = e\text{C}_\omega e$ (II_1) with normalised trace $\tau_e = \tau(\cdot)/\tau(e)$; the ZS-F23 $\mathbb{Z}_2$ seam grading β ($\beta^2 = 1$, $\tau \circ \beta = \tau$, $\beta \sigma_t = \sigma_{t\beta}$); the face algebra $\mathcal{A}_{\text{face}} = \mathbb{C}^{121}$.

§2.5. Deep-Exploration Records (v1.2–v2.0). The per-version deep-exploration records — for each version the brainstorm long-list, the MECE issue-list, the issue-tree status assignments (PROVEN / DERIVED / RETRACTED / OPEN / ...), the convergence check, and the value/posture score, as required by the exploration protocol — are collected compactly in **Appendix E** to keep the main body on the final theory. The audit trail there documents every correction and retraction made on the way to v2.0 (notably the v1.3 per-edge retraction, the v1.4→v1.5 trace/state correction, and the v1.6 scope correction).

PART I — The Dimension-Weighted Mediator Semigroup

§3. The General Connected-Graph Generator [PROVEN]

Theorem A24.I-1 (General stationary law, PROVEN). Let $G = (V, E)$ be a finite connected undirected graph with adjacency $a_{\{ij\}}$ and positive node multiplicities d_i , and define $q_{\{i \rightarrow j\}} = \Gamma_0 \kappa^2 a_{\{ij\}} d_j$ (off-diagonal), $q_{\{i \rightarrow i\}} = -\sum_{\{j \neq i\}} q_{\{i \rightarrow j\}}$. Then the unique stationary distribution is $\pi_i = d_i / \sum_j d_j$, and the chain is reversible: $\pi_i q_{\{i \rightarrow j\}} = \kappa^2 \Gamma_0 a_{\{ij\}} d_i d_j / \sum_k d_k$ is symmetric in i, j , so $\pi_i q_{\{i \rightarrow j\}} = \pi_j q_{\{j \rightarrow i\}}$. *Proof:* Appendix A; verified on the path (3, 2, 6), a 4-node mixed graph, and a triangle (4, 7, 2) — stationarity and reversibility hold for all.

This generalises A23's three-node result: the dimension-weighted rule produces the *size-by-multiplicity* equilibrium on **every** connected graph, not only the mediator path. The Z-Spin case is the path $X-Z-Y$ ($L_{XY} \equiv 0$), recovering $\pi = (3, 2, 6)/11$.

§4. Modular Detailed Balance and the Exact H-Theorem [PROVEN]

Theorem A24.I-2 (Modular detailed balance, PROVEN). With $K_i = -\ln \pi_i$, $q_{\{i \rightarrow j\}} / q_{\{j \rightarrow i\}} = d_j / d_i = \pi_j / \pi_i = e^{\{-(K_j - K_i)\}}$, i.e. $\ln(q_{\{i \rightarrow j\}} / q_{\{j \rightarrow i\}}) = -\Delta K_{\{i \rightarrow j\}}$. On the mediator path this gives $\Delta K_{\{X \rightarrow Z\}} = -\ln(3/2)$, $\Delta K_{\{Z \rightarrow Y\}} = -\ln 3$, and $\Delta K_{\{X \rightarrow Y\}} = -\ln 2$ (the ZS-F19 value) as the composite of the two edges.

Theorem A24.I-3 (Exact H-theorem, PROVEN). With detailed balance, writing $x = r_i q_{\{i \rightarrow j\}}$, $y = r_j q_{\{j \rightarrow i\}}$,

$$d/dt D(\text{rl}\pi) = -\frac{1}{2} \sum_{\{i,j\}} (x - y) \ln(x/y) \leq 0,$$

since $(x - y) \ln(x/y) \geq 0$. Hence $D(\text{rl}\pi)$ is a Lyapunov function (verified: $\max dD/dt = -7 \times 10^{-6}$ over 5000 random states; the v1.0 Monte-Carlo is retained only as a regression test). The relaxation rates are $\gamma_{\text{slow}} = 2A/Q$ and $\gamma_{\text{fast}} = A$, with $\gamma_{\text{fast}}/\gamma_{\text{slow}} = Q/2 = 11/2$, and the reverse rates $\gamma_{ZX} = 3A/Q$, $\gamma_{YZ} = 2A/Q$ (four edge rates 2 : 3 : 6 : 2). **[PREDICTION — testable on a 3-2-6 open-system simulator.]**

§5. Sector-Covariant GKLS Uniqueness [PROVEN]

Lemma A24.I-uniq (Schur per edge PROVEN; full uniqueness DERIVED; from A23.14c). Among GKLS generators that are (i) covariant under $U(d_X) \times U(d_Z) \times U(d_Y)$, (ii) free of any direct $X-Y$ channel, and (iii) primitive and trace-preserving, the jump operators on an edge (i, j) live in $\text{Hom}(H_j, H_i)$, on which $U(d_i) \times U(d_j)$ acts as $\text{fund} \boxtimes \text{antifund}$ with one-dimensional commutant; **by Schur the Kossakowski block of each edge is a scalar**, leaving *one scalar per undirected edge-orbit* — here **two** independent values γ_{XZ} and γ_{ZY} , since the $X-Z$ and $Z-Y$ Hom-spaces are inequivalent representation blocks. Covariance alone does **not** force $\gamma_{XZ} = \gamma_{ZY}$. The **global edge-homogeneity** condition — every microscopic mediator edge carries the *same* geometric impedance, the single Z-Spin constant $\kappa^2 = A/Q$ (there is one impedance, not one per sector pair) — identifies them: $\gamma_{XZ} = \gamma_{ZY} = \Gamma_0 \kappa^2$. With that condition the dimension-weighted generator is unique, realised by $L_{\{j\beta \leftarrow i\alpha\}} = \sqrt{(\Gamma_0 \kappa^2)} |j, \beta\rangle \langle i, \alpha|$ (and $L_{\{Y \leftarrow X\}} = 0$) with **no energy ladder, no equal-spacing assumption, and no bath spectral density** — $L_{XY} = 0$ reflects the absent $X-Y$ intertwiner directly. **[Sector covariance fixes one scalar per edge (Schur); the Z-Spin action-level global-edge condition identifies them with the common $\Gamma_0 \kappa^2$.]**

§6. The Full Liouvillian Spectrum — Exact and General [PROVEN, generalised in v1.5]

v1.0 computed only the 3×3 population spectrum; v1.1 reported the full 121×121 spectrum by numerical diagonalisation. **v1.5 replaces the diagonalisation with an exact representation-theoretic theorem valid for any sector dimensions**, so the 121×121 matrix audit becomes a regression test, not the proof.

Setup. Let $\dim X = a$, $\dim Z = b$, $\dim Y = c$, and set $m := a + c$, $D := a + b + c = m + b$, $\gamma := \Gamma_0 \kappa^2$. There is no direct X – Y jump, but for the Liouvillian *spectrum* the union $H_U := H_X \oplus H_Y$ ($\dim m$) couples to H_Z ($\dim b$) as a **complete bipartite** graph: every U -microstate connects to every Z -microstate with the single microscopic rate γ . The jump operators are $L_{\{z \leftarrow u\}} = \sqrt{\gamma} |z\rangle\langle u|$ and $L_{\{u \leftarrow z\}} = \sqrt{\gamma} |u\rangle\langle z|$ for $u \in U, z \in Z$.

Lemma A24.I-4a (matrix-unit action). Writing $\deg(u) = b$ (each U -state meets b Z -states) and $\deg(z) = m$, the number operator is $N := \sum_k L_k^\dagger L_k = \gamma(b \cdot P_U + m \cdot P_Z)$. For an off-diagonal unit $E_{rs} = |r\rangle\langle s|$ ($r \neq s$) the recycling term $L_k E_{rs} L_k^\dagger$ vanishes, so

$$\mathcal{L}(E_{rs}) = -(\gamma/2)(\deg r + \deg s) E_{rs}.$$

Hence E_{rs} is already an eigenvector. *Proof:* the recycling term $\gamma\langle\alpha|r\rangle\langle s|\alpha\rangle|\beta\rangle\langle\beta|$ is nonzero only if $r = s = \alpha$.

Invariant decomposition. This yields four off-diagonal eigenspaces and a diagonal block:

Operator class	Eigenvalue	Count
U – U coherence E_{uv} ($u \neq v$)	$-b\gamma$	$m(m-1)$
Z – Z coherence E_{zw} ($z \neq w$)	$-m\gamma$	$b(b-1)$
U – Z and Z – U coherence E_{uz}, E_{zu}	$-D\gamma/2$	$2mb$
diagonal $D_U^0 = \{\sum x_u E_{uu} : \sum x_u = 0\}$	$-b\gamma$	$m-1$
diagonal $D_Z^0 = \{\sum y_z E_{zz} : \sum y_z = 0\}$	$-m\gamma$	$b-1$
diagonal span $\{I_U, I_Z\}$	$0, -D\gamma$	2

The recycling term cancels on D_U^0 and D_Z^0 (because every U -state meets the *same* Z -set and $\sum x_u = 0$), giving the bare decay there; on span $\{I_U, I_Z\}$ the 2×2 block $\begin{bmatrix} -b\gamma & b\gamma \\ m\gamma & -m\gamma \end{bmatrix}$ has eigenvalues 0 (trace $= \pi$) and $-D\gamma$.

Theorem A24.I-4G (General exact Liouvillian spectrum, PROVEN). Collecting the U – U coherences with D_U^0 (total m^2-1 at $-b\gamma$) and the Z – Z coherences with D_Z^0 (total b^2-1 at $-m\gamma$),

$$\text{Spec}(\mathcal{L}) = \{ 0^{\wedge\{1\}}, (-b\gamma)^{\wedge\{m^2-1\}}, (-D\gamma/2)^{\wedge\{2mb\}}, (-m\gamma)^{\wedge\{b^2-1\}}, (-D\gamma)^{\wedge\{1\}} \},$$

with characteristic polynomial $\chi_{\mathcal{L}}(\lambda) = \lambda(\lambda + D\gamma)(\lambda + b\gamma)^{\wedge\{m^2-1\}}(\lambda + D\gamma/2)^{\wedge\{2mb\}}(\lambda + m\gamma)^{\wedge\{b^2-1\}}$ and dimension sum $1 + (m^2-1) + 2mb + (b^2-1) + 1 = (m + b)^2 = D^2$. *Proof:* Lemma A24.I-4a plus the invariant decomposition above; verified against the explicit 121×121 build for $(3, 2, 6)$ and against a second case $(2, 3, 4)$ in `zs_a24_v2_1_audit.py`.

Corollary A24.I-4 (Z-Spin, PROVEN). For $(a, b, c) = (3, 2, 6)$, $m = 9$, $D = 11$, $\gamma = \kappa^2 = A/Q$ (so $D\kappa^2 = A$),

$$\text{Spec}(\mathcal{L}) = \{ 0, -2A/Q, -A/2, -9A/Q, -A \}, \text{ multiplicities } (1, 80, 36, 3, 1), \Sigma = 121.$$

The 36-fold $-A/2$ block is the $X-Z$ and $Z-Y$ cross-sector coherence space ($6 + 6$ and $12 + 12$), decohering at exactly half the fast population rate A . Every relaxation and decoherence rate is a rational multiple of A , with **no new parameters**. The earlier numerical diagonalisation is now a regression check, not the derivation.

§7. Perturbation Stability [two theorems PROVEN in v1.5; generic numerical]

Real hardware never realises perfectly uniform jumps. v1.1 bounded the response numerically; v1.5 splits the perturbation into two MECE classes and closes each with an exact theorem.

§7.1 Reciprocal (detailed-balance-preserving) perturbations [PROVEN]. Write the reversible conductance $C_{ij} := \pi_i q_{ij} = C_{ji}$, and perturb the *conductance* symmetrically: $C_{ij}(\epsilon) = C_{ij}(1 + s_{ij})$, $s_{ij} = s_{ji}$, $|s_{ij}| \leq \epsilon < 1$, with $q_{ij}(\epsilon) = C_{ij}(\epsilon)/\pi_i$. Then π is **exactly** unchanged and detailed balance is exactly preserved ($\pi_i q_{ij}(\epsilon) = C_{ij}(\epsilon) = C_{ji}(\epsilon) = \pi_j q_{ji}(\epsilon)$). The Dirichlet form $E_\epsilon(f, f) = \frac{1}{2} \sum_{ij} C_{ij}(1 + s_{ij})(f_i - f_j)^2$ obeys $(1 - \epsilon)E_0 \leq E_\epsilon \leq (1 + \epsilon)E_0$, so by Rayleigh–Ritz

$$(1 - \epsilon) \lambda_{\text{gap}} \leq \lambda_{\text{gap}}(\epsilon) \leq (1 + \epsilon) \lambda_{\text{gap}}, \quad \lambda_{\text{gap}} = 2A/Q.$$

The same bound holds for the full GKLS Liouvillian (each undirected microscopic pair carries $\gamma_{uz} = \gamma(1 + s_{uz})$, $s_{uz} = s_{zu}$; the HS Dirichlet form is a nonnegative sum and scales identically).

[PROVEN for all $0 \leq \epsilon < 1$.]

§7.2 Directional (3-state path) perturbations [PROVEN]. Let the four independent rates be $\alpha = q_{X \rightarrow Z}$, $\beta = q_{Z \rightarrow X}$, $\delta = q_{Z \rightarrow Y}$, $\eta = q_{Y \rightarrow Z}$. The matrix-tree theorem gives the **exact** stationary measure

$$\pi^\epsilon = (\beta\eta, \alpha\eta, \alpha\delta) / (\beta\eta + \alpha\eta + \alpha\delta),$$

and the two nonzero decay rates are exactly $g_\pm = (S \pm \sqrt{S^2 - 4T})/2$ with $S = \alpha + \beta + \delta + \eta$, $T = \alpha\delta + \alpha\eta + \beta\eta$. At the base rates ($\alpha_0 = \eta_0 = b\gamma$, $\beta_0 = a\gamma$, $\delta_0 = c\gamma$) this returns $\pi = (3, 2, 6)/11$, $g_- = b\gamma = 2A/Q$, $g_+ = D\gamma = A$. With $r_k = r_{k,0}(1 + \epsilon_k)$, $|\epsilon_k| \leq \epsilon$, each spanning-tree weight is a product of two rates, giving the rigorous total-variation bound $\|\pi_\epsilon - \pi\|_{\text{TV}} \leq 2\epsilon/(1 + \epsilon^2)$. The first-order gap shift is

$$g_-^{-\epsilon} = b\gamma[1 + (c\epsilon_{X \rightarrow Z} + a\epsilon_{Y \rightarrow Z})/(a + c)] + O(\epsilon^2)$$

— it depends on the **forward** rates only; the reverse rates $\epsilon_{Z \rightarrow X}$, $\epsilon_{Z \rightarrow Y}$ drop out at first order (a new analytic result, invisible to the earlier Monte Carlo). For $(a, c) = (3, 6)$: $g_-^{-\epsilon} = 2\gamma[1 + (2/3)\epsilon_{X \rightarrow Z} + (1/3)\epsilon_{Y \rightarrow Z}]$. [PROVEN.]

§7.3 Generic perturbations — coarse and full microscopic [VERIFIED-NUMERICAL]. Two distinct Monte-Carlo tests. (a) **Coarse 3-state generic (non-reciprocal) perturbation:** the four population rates of the 3-state chain are shaken independently (500 random perturbations per scale):

ϵ (rate)	gap (base $2A/Q = 0.01456$)	$\max_i \Delta\pi_i $
± 0.01	0.01456 ± 0.00006	0.0065
± 0.05	0.01456 ± 0.00032	0.0315
± 0.10	0.01452 ± 0.00064	0.0599
± 0.20	0.01459 ± 0.00127	0.1138

(b) **Full microscopic 121×121 symmetry-breaking:** each of the 36 microscopic jump *rates* is perturbed independently and the full 121×121 superoperator is rebuilt every trial; at ±5% the spectral gap deviates by ≤ 5% and the sector populations by $\max_i |\Delta p_i| < 0.02$ (the audit's PASS threshold; the fixed-seed run gives ≈ 0.011), consistent with the §7.1–7.2 analytic bounds. **[VERIFIED-NUMERICAL — both the coarse chain and the full 121×121 microscopic Liouvillian are MC-tested (zs_a24_v2_1_audit.py); the figure is a robustness illustration, not a certified statistical tolerance.]**

§8. Simulator Protocol and Hardware Tolerance [TESTABLE]

The architecture $\mathbf{H} = \mathbb{C}^3 \oplus \mathbb{C}^2 \oplus \mathbb{C}^6$ (a 3-level X register, a 2-channel Z mediator, a 6-channel Y reservoir; ZS-QH) realises both generators by a choice of jump amplitude. The two stationary samplings are **distinct dynamics** and require **two protocols** (π is the stationary state of $\mathbf{L}_\pi$, ω that of $\mathbf{L}_\omega$):

Protocol A (state sampling). Activate only the X–Z and Z–Y channels with *equal* microtransition amplitude $\sqrt{(\Gamma_{\text{ok}})^2}$ (this is $\mathbf{L}_\pi$). Measure $p_X(t)$, $p_Z(t)$, $p_Y(t)$. **Predictions:** $p(\infty) = (3, 2, 6)/11 = \pi$; $\lambda_{\text{slow}} = -2\mathbf{A}/\mathbf{Q}$, $\lambda_{\text{fast}} = -\mathbf{A}$, $\lambda_{\text{fast}}/\lambda_{\text{slow}} = \mathbf{Q}/2 = 11/2$; cross-sector coherences decay at $\mathbf{A}/2$ (a Ramsey/echo measurement).

Protocol B (operator sampling). Tilt each jump amplitude by $(d_j/d_i)^{1/4}$ (this is $\mathbf{L}_\omega$). Then the stationary populations are $p(\infty) = (9, 4, 36)/49 = \omega$; observer tomography of the operator-size-biased weight returns ω .

Tolerance (§7): ≤ 5% **rate** non-uniformity (≈ 2.5% amplitude, since $\text{rate} \propto \text{amplitude}^2$) holds the stationary populations to an absolute deviation $\max_i |\Delta p_i| \lesssim 0.03$ (the §7.3 figure is an absolute probability deviation, not a ratio error). **[TESTABLE — the entire Part I/Part III finite-register dynamics is verifiable on a small open-system simulator without awaiting any cosmological scale.]**

PART II — The Spin-Graded Continuous-Core Lift

§9. Gate 1, Step 0 — Why the Finite Route Fails [CLOSED-NEGATIVE]

Theorem A24.II-0 (Finite-route obstruction, CLOSED-NEGATIVE). Let $\sigma_t^\omega = \text{Ad}(\rho_\omega^{\text{it}})$ be the modular flow of ω_{reg} on M_{11} . Then:

1. **σ^ω is inner.** M_{11} is a finite-dimensional factor; by Skolem–Noether every automorphism is inner, and σ_t^ω is implemented by $\rho_\omega^{\text{it}} \in M_{11}$.
2. **The crossed product is not a II factor.** For an inner one-parameter action implemented by $U(t) \in M$, the crossed product $M \rtimes_{\{\text{Ad } U\}} \mathbb{R}$ is isomorphic to $M \rtimes_{W^*(U(t))} \mathbb{R} \cong M_{11} \rtimes_{L^\infty(\mathbb{R})} \mathbb{R}$, with diffuse centre $1 \otimes L^\infty(\mathbb{R})$. This is a Type I algebra with continuous centre — **not a Type II factor**, and it carries no canonical finite trace distinct from the M_{11} trace tensored with Lebesgue measure.

Hence v1.0's Theorem A24.1(a) is **RETRACTED** for the finite route, and the v1.0 statements "This seals Gate 1" and "the Type II trace recovers ..." are withdrawn. The error in v1.0 was to read the *non-triviality* of σ^ω (the d_i differ) as *outerness*; a non-trivial inner action is still inner. *Verified:* audit II.1.

§10. The Genuine Type III $\rightarrow$ II $_{\infty}$ Continuous Core [IMPORTED-PROVEN]

Theorem A24.II-1 (Continuous core, IMPORTED-PROVEN). Let $\mathcal{A}^{\text{III}}$ be the Type III $_1$ von Neumann algebra of quantum-field observables in a de Sitter static patch (CLPW). Its Tomita–Takesaki modular automorphism group σ_t (for the cyclic separating vacuum) is **outer**. The crossed product

$$\mathbf{C}_{\omega} := \mathcal{A}^{\text{III}} \rtimes_{\sigma} \mathbb{R}$$

is a **Type II $_{\infty}$** factor carrying a faithful normal semifinite **canonical trace** τ — distinct from the dual weight, which is $\varphi(x) = \tau(h_{\varphi} x)$ for the modular density h_{φ} (the same trace-vs-weight distinction made precise for the finite register in §12). Takesaki's continuous decomposition of a Type III $_1$ factor, specialised to gravity by Witten ("gravity and the crossed product") and to the static patch by CLPW, supplies it. The gauged modular boost intrinsically regularises the entanglement entropy with no explicit UV cutoff; in Z-Spin the polyhedral lattice supplies the finite-dimensional regulator (ZS-F19 §12). The $\mathbb{Z}_2$ seam grading (ZS-F23) acts as a trace-preserving order-2 automorphism β ($\beta^2 = 1$, $\tau \circ \beta = \tau$, $\beta \sigma_t = \sigma_{t\beta}$); the spin-graded core is $\mathbf{C}_{\omega} \rtimes \beta$. *This theorem is imported, not re-proved.*

The construction direction is therefore $\mathcal{A}^{\text{III}} \rtimes_{\sigma} \mathbb{R} \rightarrow \mathbf{C}_{\omega} \text{ (II}_{\infty}\text{)} \xrightarrow{\text{--corner--}} \mathbf{M}_{\text{obs}} \text{ (II}_1\text{)}$, into which the finite register embeds $\mathbf{M}_{11} \xrightarrow{\text{--}\iota\text{--}} \mathbf{M}_{\text{obs}} \subset \mathbf{C}_{\omega}$, not the (false) " $\mathbf{M}_{11} \xrightarrow{\text{--cross--}} \text{II}.$ "

§11. The Finite II $_1$ Corner and the Register Embedding [DERIVED-CONDITIONAL]

Theorem A24.II-2 (Finite corner and embedding, DERIVED-CONDITIONAL). Choose a projection $e \in \mathbf{C}_{\omega}$ with $0 < \tau(e) < \infty$. Then the corner

$$\mathbf{M}_{\text{obs}} := e \mathbf{C}_{\omega} e \text{ is a Type II}_1 \text{ factor, with normalised trace } \tau_e(x) = \tau(exe)/\tau(e),$$

(a finite corner of a II $_{\infty}$ factor is II $_1$). The finite register embeds as a unital $*$ -homomorphism

$$\iota : \mathbf{M}_{11} \hookrightarrow \mathbf{M}_{\text{obs}},$$

and the A23 dimension-weighted semigroup acts on $\iota(\mathbf{M}_{11})$ as the population dynamics (Part I), so that a conditional expectation $E : \mathbf{M}_{\text{obs}} \rightarrow \iota(\mathbf{M}_{11})$ carries it to the modular clock of $\mathbf{C}_{\omega}$ at the **population** level. Lifting the full **bridge** to the core *would require* a generator $\mathcal{L}_s$ on $\mathbf{M}_{\text{obs}}$ (yet to be constructed) intertwining the finite generator $\mathbf{L}_s$ built on $\mathbf{M}_{11}$ in §U4,

$$E \circ \mathcal{L}_s = \mathbf{L}_s \circ E \text{ (with modular covariance and complete positivity),}$$

which is **not** done here. **Conditional on:** the existence of the embedding ι — proved *realisable* (II $_1$ factors admit such unital embeddings, for which the trace is automatically the normalised matrix trace) but not *selected from the Z-Spin action*. The remaining conditions are not the trace weight (that is fixed) but the **compatibility of the modular density h with the core dual weight** and the **dynamical lift $\mathcal{L}_s$** (§U6, OPEN). [DERIVED-CONDITIONAL on the existence of ι and modular-density compatibility; the trace-preserving conditional expectation E exists, but the dynamical intertwining $E \circ \mathcal{L}_s = \mathbf{L}_s \circ E$ is a requirement on the still-unconstructed continuous-core lift and remains OPEN (F-A24.9).]

§12. Gate 2 — Observer Trace [corrected v1.5: trace = π , observer weight $\omega = \tau(h \cdot)$] and CLPW Offset [Gate 2(ii) DERIVED-CONDITIONAL]

Theorem A24.II-3 (Observer trace and observer state, DERIVED — corrected in v1.5). The II $_1$ -corner **trace** of the embedded projectors is the **matrix trace**

$$\tau_e(u(P_i)) = d_i/D = (3, 2, 6)/11 = \pi \text{ (the Part-I stationary),}$$

realised in continuous trace dimension (each of the **11 Hilbert microstate projections** e_α has $\tau_e(e_\alpha) = 1/D = 1/11$, well-posed in Π_1 where the v1.0 rank statement was not; the symbol q_α is reserved for the **121 operator-slot** projections of §13, $\tau_e(q_\alpha) = 1/121$). The A23 **observer weight** $(9, 4, 36)/49$ is **not this trace** but the **state**

$$\omega_e(x) = \tau_e(h x), \quad \omega_e(u(P_i)) = d_i^2/\Sigma_j d_j^2 = (9, 4, 36)/49, \quad h_i = D d_i/\Sigma_j d_j^2,$$

where h is the modular density ($\tau_e(h) = 1$). Both are PROVEN: $\tau_e(u(P_i)) = d_i/11$ is the matrix trace, and $\omega_e(u(P_i)) = \tau_e(h u(P_i)) = (1/11)(11 d_i/49) d_i = d_i^2/49$.

v1.5 correction (this fixes a conflation carried from v1.0). v1.4 identified the corner trace itself with $(9, 4, 36)/49$ via the Takesaki dual weight. That **conflated a state with a trace**: the Takesaki dual weight of $C_\omega = \mathcal{A}^{\text{III}} \rtimes_\sigma \mathbb{R}$ is the **trace times the modular density** ($\varphi = \tau(h_\varphi \cdot)$), so restricting it to the register gives the *state* $\omega_{\text{reg}} = \tau_e(h \cdot)$, **not** the trace. The corner *trace* is the matrix trace, $\tau_e(u(P_i)) = d_i/11 = \pi$. This also resolves an internal inconsistency: v1.4's own audit check II.2b already recorded that the rank-normalised corner trace is $(3, 2, 6)/11$; and it **restores consistency with ZS-A23**, which calls $(3, 2, 6)/11$ the "trace weights / stationary distribution" (A23.5a) and $(9, 4, 36)/49$ the operator-sampling state ω_{reg} (A23.MC). ZS-A23.MC and A23 §5(iii) PROVE $\omega_i = d_i \pi_i/\Sigma_j d_j \pi_j = d_i^2/\Sigma_j d_j^2$ two independent ways (the operator-space size-bias of π , and the modular centralizer of ω_{reg} with density $h_i = 11 d_i/49$ — exactly the h above).

Consequence — the unification. Because the corner carries the *same* trace π as the Part-I dynamics, and ω is the modular state $\tau_e(h \cdot)$ on the *same* algebra, the dimension-weighted L_π and the observer-balanced L_ω are two reversible dynamics on one algebra joined by a modular half-density transform — the **finite-register** Part I $\leftrightarrow$ Part II unification developed in **PART III** (the continuous-core dynamical lift remains OPEN, F-A24.9). [**Gate 2(i): DERIVED — corner trace = matrix trace = $\pi = (3, 2, 6)/11$; observer weight $\omega = (9, 4, 36)/49 = \tau_e(h \cdot)$, the modular state. The v1.4 trace-vs-state conflation is corrected.**]

Theorem A24.II-4 (CLPW offset, IMPORTED-MATCHING / DERIVED-CONDITIONAL). CLPW give $S_{\text{alg}} = S_{\text{gen}} + C$ with C a state-independent additive constant fixed by the Type II renormalisation; the maximum-entropy state of the Π_1 corner is empty de Sitter. Choosing empty de Sitter as the normalised tracial *reference* (where $S_\tau = 0$) matches the offset to the de Sitter horizon entropy,

$$S_0 = A_{\text{dS}}/4G = 8\pi^2 \bar{M}_P^2/H^2 \quad (\text{Gibbons–Hawking}).$$

This is a **matching under a chosen reference normalisation, not a unique algebraic derivation** — the corpus (Book/ZS-F23) keeps the generalised-entropy formula itself unproved, the identification of a specific hyperfinite factor with the gravitational subregion algebra as Condition C (OPEN), and the observer/clock identification as HYPOTHESIS-level. v1.0 over-stated this; v1.1 lowers it. The residual $O(1)$ and the absolute scale reduce to the corpus-wide **B3** scale $\bar{H}/\bar{M}_P$ (§15). [**Gate 2(ii): IMPORTED-MATCHING / DERIVED-CONDITIONAL.**]

§13. Gate 3 — The Face Embedding: Partition DERIVED from ZS-F2 §11.4 [DERIVED-CONDITIONAL]

v1.0 wrote that $\mathcal{A}_{\text{face}} = \mathbb{C}^{121}$ embeds "as a MASA of M_{11} ." This is **mathematically false**: a maximal abelian subalgebra of M_{11} is $\mathbb{C}^{11}$ (the diagonal), there is no unital injective $*$ -homomorphism $\mathbb{C}^{121} \hookrightarrow M_{11}$ (one cannot place 121 orthogonal nonzero projections in M_{11}), and (6, 32, 83) cannot be projection ranks in M_{11} (they sum to $121 > 11$). The error was to confuse the operator dimension (121) with a projection count. The correct target lives in the **Π_1 corner**, where projections carry *continuous trace* $\in [0, 1]$, not integer rank.

Corrected Gate-3 target (well-posed, OPEN). The face embedding is a trace-preserving $*$ -homomorphism

$$\Phi_{\text{face}} : \mathcal{A}_{\text{face}} = \mathbb{C}^{121} \rightarrow M_{\text{obs}}(\Pi_1), \quad e_a \mapsto q_a, \quad q_a q_b = \delta_{ab} q_a, \quad \tau_e(q_a) = 1/121,$$

sending the 121 minimal face idempotents to 121 mutually orthogonal projections each of continuous trace $1/121$ (possible in a Π_1 factor; impossible in M_{11}). The coarse projections

$$q_b = \sum_{a \in B} q_a, \quad q_c = \sum_{a \in C} q_a, \quad q_\Lambda = \sum_{a \in \Lambda} q_a, \quad \text{with } \tau_e(q_b, q_c, q_\Lambda) = (6, 32, 83)/121,$$

then carry the face-counting budget $\Omega_b = 6/121$, $\Omega_{\text{cdm}} = 32/121$, $\Omega_\Lambda = 83/121$ (ZS-F2; Planck 0.4%) as **trace dimensions, not ranks**.

v1.2 — the partition is DERIVED from ZS-F2 §11.4. The three-block decomposition is no longer asserted; it is read off the face-counting construction (ZS-F2 §11.4, Theorem 11.7, Boundary Mode Theorem):

$b = F(\text{cube}) = 6$ (the X-sector spatial frame; visible matter = $F(\text{cube})/Q^2 = 6/121 = \text{slot count } X \cdot Z/Q^2$);

$c = F(\text{truncated icosahedron}) = 32$ (the $Y \rightarrow X$ mediator; $\text{CDM} = \dim \Omega^2(\Gamma_{\text{med}}) = F(\Gamma_{\text{med}}) = 32/121$, the Hodge 2-form dimension);

$$\Lambda = 121 - 6 - 32 = 83 \text{ (the remaining background slots; } \Omega_\Lambda = 83/121),$$

with $Q^2 = 121 = \dim M_{11}$ the operator dimension and one slot = $1/Q^2$ per face. ZS-F2's anti-numerology already certifies this: of $n \in [1, 120]$ only $n = 32$ places ω_{cdm} within 1σ of Planck, 0.0 % of 500 000 random fractions outperform $32/121$, and $32 = F(\text{truncated icosahedron})$ was the Y-mediator assignment *before* the face-counting discovery. The A24 contribution is to realise ZS-F2's "121 operator slots" as **121 genuine orthogonal projections of continuous trace $1/121$ in the Π_1 corner** — which M_{11} cannot host but a Π_1 factor can — so that (6, 32, 83)/121 are well-defined trace dimensions. **[Gate-3 partition: DERIVED (ZS-F2 §11.4); realised as Π_1 projections (A24).]**

What remains conditional. The equal one-quantum-per-face weight $\tau_e(q_a) = 1/121$ is **combinatorial (not metric) holography** — the polyhedral faces have unequal areas (cube ≈ 1.33 , dodecahedron ≈ 0.88), so a metric/area weighting would be unequal. **v1.3 resolves the combinatorial-vs-metric question in favour of combinatorial.** ZS-F0 §8.5 (Theorem 8.9) constructs the Z-sector partition function as a **BV–BFV cobordism invariant**: the closed 4-cobordism Wilson loop W maps under the functor B_Z to the i -tetration evolution, with partition function $Z(W) = (i\pi/2)z \setminus \text{and survival probability } |Z(W)|^2 = (\pi^2/4)\eta_{\text{topo}} \approx 0.7948$ — a **topological quantity (the i -tetration trace)**, not a metric/area integral. *A topological boundary partition function counts states combinatorially (one quantum per face,*

area-independent), so the equal weight $1/121$ is the combinatorial dimension count, **not** an area weight. The residual is therefore narrowed to a single deferred item: the explicit face-weight state-sum (a Turaev–Viro / Barrett–Westbury realisation with Pachner-move invariance forcing equal weights), which is ZS-F0's own deferred categorical formalisation (F0 §166, "DERIVED-CONDITIONAL ... modulo categorical polish"). **v1.4 — the equal weight is the monoidal functor's per-face assignment; the residual is upstream.** ZS-F0 §8.3 (Definition 8.3) makes B_Z a **monoidal functor** that assigns the sector decomposition of $\mathbb{C}^{11}$ to each boundary cell. By functoriality and locality, every face of the same type receives the same assignment (one copy of the representation space per face), so the weight is equal per face — the combinatorial $1/121$. The explicit* face-weight state-sum (the Turaev–Viro / Barrett–Westbury realisation with Pachner-move invariance) is precisely **ZS-F0's own deferred categorical formalisation** (F0 §166: "Full closure requires the formalization of the BV-BFV monoidal functor structure at the categorical level"). It is therefore an **upstream** item — ZS-F0's to complete — not an A24-internal gap. **[Gate 3: partition DERIVED (ZS-F2 §11.4); combinatorial nature DERIVED (ZS-F0 §8.5); equal weight = the local monoidal functor's per-face assignment (ZS-F0 §8.3); the only residual is ZS-F0's deferred categorical state-sum (UPSTREAM), not A24.]**

§14. Gate 4 — The Two-Edge Instanton: Factor 2 DERIVED, Per-Edge Corrected to π/A (F-A24.4)

A23 sharpened the period-2 absolute factor $|\Delta W| = 2$ to a two-edge structure and left the per-edge action OPEN. The reviewer correctly noted that two-edge additivity $S_ \{X \rightarrow Y\} = S_ \{X \rightarrow Z\} + S_ \{Z \rightarrow Y\}$ alone gives neither equal edges nor the value $1/\kappa^2$. F-A24.4 resolves the equality and reports the value honestly.

§14.1 The factor 2 is DERIVED [equal edges].

Theorem A24.II-5 (Equal-edge doubling, DERIVED). Both seam crossings of the path $X-Z-Y$ are mediated by the **same** Z-sector topological object — the Z-Anchor Bogomolnyi vortex, the unique $|\Phi| = 0$ core forced by $\pi_1(U(1)) = \mathbb{Z}$ (ZS-F1). The instanton action of a BPS vortex is *topological*: it is fixed by the winding number and the VEV, and is **independent of the sector multiplicities d_i** (which weight the dissipative rate $q = \kappa^2 d_j$, a logically separate structure). Hence $S_ \{X \rightarrow Z\} = S_ \{Z \rightarrow Y\} = S_ \text{vortex}$ despite the different edge dimensions ($3 \leftrightarrow 2$ vs $2 \leftrightarrow 6$), and

$$S_ \{X \rightarrow Y\} = 2 S_ \text{vortex}.$$

This upgrades the period-2 factor from "graph-distance additivity" to **exact doubling**: the cosmologically relevant factor $|\Delta W| = 2$ is DERIVED. **[Gate 4 factor 2: DERIVED.]**

§14.2 The per-edge value: $1/\kappa^2$ RETRACTED, corrected to the seam-flip π/A [F-A24.4, v1.3].

Computation A24.II-6 (the Z-Anchor action from ZS-F1, v1.2). Reading the actual ZS-F1 action settles what kind of object the Z-Anchor is. ZS-F1 §7 keeps the Goldstone θ **massless and physical** ($\Delta N_ \text{eff} = 0$; it drives the ZS-A1 isothermal halo), so the $U(1)$ is **global**, not Higgsed. A global vortex has a **logarithmically divergent** energy $E \approx \pi v^2 \ln(L/\xi)$ ($\xi = \hbar c/m_p \approx 31 \text{ l}_P$, ZS-F1 §5.3), **not** the BPS-finite tension $2\pi v^2 |n|$. Two readings, the same conclusion:

- **Global (the ZS-F1 reading):** per-edge action $\propto \ln(L/\xi)$, IR-cutoff-dependent — not a clean $1/\kappa^2$ at all.

- **Gauged BPS (counterfactual):** $S_{\text{vortex}} = 2\pi v^2 |n| = 2\pi \approx 6.28$ for the minimal $v = 1$, $n = 1$ — and $S_{\text{vortex}} = 1/\kappa^2 = Q/A \approx 137.34$ would require $v^2 |n| = Q/(2\pi A) \approx 21.86$, i.e. winding $n \approx 22$, which ZS-F1 §5 **excludes** (the minimal topologically-stable winding is $n = 1$).

In neither reading does the Z-Anchor supply per-edge = $1/\kappa^2$. **The Bogomolnyi-vortex route to the per-edge action is therefore CLOSED-NEGATIVE.**

Computation A24.II-7 (the θ -instanton search and the corrected per-edge value, v1.3). v1.2 left per-edge = $1/\kappa^2$ to a Z-EFT θ -instanton. Reading the corpus instanton ledger settles it: the corpus's *actual* non-perturbative actions are $t_{\text{flip}} = \pi/A \approx 39.22$ (one $\mathbb{Z}_2$ half-event, ZS-M3 §6), $T_{\text{micro}} = 2\pi/A \approx 78.45$ (one full period), $S_{\text{tunnel}} = 5\pi/A \approx 196$ (five seam flips, ZS-M3/ZS-A3 proton decay), and $S_{\text{cl}} = 35\pi/3 \approx 36.65$ (the EWSB instanton, ZS-S4 §6.11). **None equals $1/\kappa^2 = Q/A \approx 137.34$** — indeed $1/\kappa^2 \div (\pi/A) = Q/\pi \approx 3.50$ is not an integer, so $1/\kappa^2$ is not a seam-flip count; a θ -instanton tuned to $1/\kappa^2$ would *conflict* with these established scales (exactly the conflict that demotes the ZS-M14 NC-4 $1/\alpha$ -instanton term to HYPOTHESIS-weak). **So per-edge = $1/\kappa^2$ is RETRACTED.** Since the A24 mediation **crosses the $\mathbb{Z}_2$ seam** (ZS-F23: X-even $\rightarrow$ Y-odd through the Z-seam, no direct X–Y channel), its per-edge action is the corpus seam-flip:

$$\text{per-edge action} = t_{\text{flip}} = \pi/A \text{ (one } \mathbb{Z}_2 \text{ half-event, ZS-M3 §6, DERIVED), } S_{\{X \rightarrow Y\}} = 2 \times (\pi/A) = 2\pi/A = T_{\text{micro}},$$

with the **factor 2 = the two $\mathbb{Z}_2$ half-events of one full seam event** — exactly v1.2's "two equal edges = the same topological Z-Anchor," now identified with ZS-M3's $T_{\text{micro}} = 2 t_{\text{flip}}$. **[Gate 4 per-edge value: $1/\kappa^2$ RETRACTED; per-edge = π/A DERIVED-CONDITIONAL on the A24-edge = ZS-M3-half-event identification; the Z-EFT θ -instanton route is superseded — the corpus seam action is π/A , not $1/\kappa^2$.]**

The $1/\kappa^2 \leftrightarrow \Lambda$ motivation was numerology. A23 chose per-edge = $1/\kappa^2$ because $S_{\{X \rightarrow Y\}} = 2/\kappa^2$ gives $e^{\{-2/\kappa^2\}} \approx 5 \times 10^{-120} \approx (\bar{H}_0/\bar{M}_P)^2$ — a *fit* to the cosmological-constant scale. The corpus-grounded value gives $e^{\{-2\pi/A\}} \approx 8 \times 10^{-35}$, which is **not Λ** ; and ZS-F1 §6.4 derives Λ **geometrically** (the $(1 + A)$ modification with $V(1) = 0$). The $e^{\{-2/\kappa^2\}} \approx \Lambda$ coincidence is therefore an **anti-numerology RETRACTION**: it played no derivational role, the true seam action does not reproduce it, and Λ is geometric.

v1.4 — the same-seam lemma fixes the scale. The identification "A24 per-edge = a ZS-M3 seam-flip" rests on one fact: the A24 seam *is* the ZS-M3 seam. ZS-F23's $\mathbb{Z}_2$ seam grading is the J-grading of ZS-F0 §8.4 (Theorem 8.5–8.6: $|Jj\rangle = |10-j\rangle$), the unique fixed point $|5\rangle$, forced by $Q = 11$ odd), and ZS-M3's Z-Telomere flips exactly this $\mathbb{Z}_2$. They are **one object**, so the A24 mediation's seam crossing carries the ZS-M3 seam-flip action. ZS-M3 §8's hierarchy $\tau_n = t_P e^{\{n\pi/A\}}$ establishes **π/A as the fundamental per-flip action unit**; hence the per-edge action is fixed *at the π/A scale* (DERIVED), with per-edge = $t_{\text{flip}} = \pi/A$ and $S_{\{X \rightarrow Y\}} = T_{\text{micro}} = 2\pi/A$ on the two-edge = two-half-event reading. The only residual is the leg-count (whether one A24 edge is one $\mathbb{Z}_2$ half-event or half of one — i.e. per-edge = π/A vs $\pi/2A$), which is **non-load-bearing**: the cosmological factor 2 and the π/A scale are DERIVED regardless, and nothing downstream depends on the exact multiple. **[Gate 4 per-edge: scale π/A DERIVED (same-seam lemma + ZS-M3 §8); the exact value (π/A) DERIVED-CONDITIONAL on the leg-count, which is non-load-bearing.]**

§15. The Physics Payoff — B3 Absolute Scale and the C5 Clock-DOF Candidate [OPEN]

§15.1 B3 stays OPEN. The Π_1 trace carries precisely a state-independent additive constant (CLPW), which is *exactly* the freedom that leaves the absolute scale $\bar{H}/\bar{M}_P \approx 5.9 \times 10^{-61}$ undetermined. No action-level mechanism fixes it; the DERIVED hierarchy $\tau_n = t_P \cdot \exp(n\pi/A)$ (ZS-F10/U8) spans the range but does not pin $\bar{H}_0$. **v1.3 note (anti-numerology RETRACTION).** v1.2 recorded, as an OBSERVATION, the hope that the per-edge instanton feeds B3 — *if* per-edge = $1/\kappa^2$ then $e^{\{-2/\kappa^2\}} \approx 5 \times 10^{-120}$, within ~ 2 orders of $(\bar{H}_0/\bar{M}_P)^2 \approx 3 \times 10^{-121}$. **v1.3 retracts this:** §14.2 shows the corpus seam action is π/A , not $1/\kappa^2$, giving $e^{\{-2\pi/A\}} \approx 8 \times 10^{-35}$ — nowhere near Λ . The $e^{\{-2/\kappa^2\}} \approx \Lambda$ match was a numerical coincidence with no derivational role, and the true seam action does not reproduce it. ZS-F1 §6.4 derives Λ **geometrically** (the $(1 + A)$ modification with $V(1) = 0$). So B3 is **not** fed by the per-edge action and is **not** built on any $e^{\{-2/\kappa^2\}}$ coincidence. **A24 does not close B3.** [OPEN; NON-CLAIM; the v1.2 per-edge $\leftrightarrow$ B3 OBSERVATION is RETRACTED.]

§15.2 The clock DOF as a C5 candidate [HYPOTHESIS-weak]. The crossed product $\rtimes_{\sigma} \mathbb{R}$ of Part II adjoins one continuous degree of freedom — the clock — which is structurally the Quintom ingredient (i) ("an extra propagating field") that A22 (Prop A22.8) requires for a healthy $w = -1$ crossing. This registers the clock DOF as a C5 candidate. The barrier is A22's B1 (reflection-positivity / healthiness): the clock-sector EFT must be shown to propagate healthily in the IR and to satisfy OS-3. Neither is shown here. **The continuous-core lift opens, rather than closes, the C5 audit; a confirmed DESI crossing would activate it.** [HYPOTHESIS-weak; closure route = a reflection-positivity proof for the clock-sector EFT; failing chain = OS-3 violation (ZS-M17.3).]

PART III — The Modular Half-Density Bridge: Finite-Register Unification (and the Conditional Lift)

Part I built a reversible dynamics L_π whose stationary distribution is the state-sampling weight $\pi \propto d_i$; Part II identified the observer weight $\omega \propto d_i^2$. The two-Part split left open whether these are one structure. **At the finite-register level they are — not as one stationary state, but as two reversible dynamics on the same algebra M_{11} joined by a modular half-density transform.** This Part proves the finite-register statement (and proves L_ω is a genuine quantum-detailed-balance generator), then states precisely what remains to lift it to the continuous core (§U6, OPEN).

§U1. The no-go (why a single generator does not suffice) [PROVEN]. The Part-I coarse generator is $q_{\{i \rightarrow j\}} = \gamma d_j$, with stationary $\pi_i = d_i/D$. Suppose the *same* generator were ω -detailed-balance. Then $\omega_i q_{\{i \rightarrow j\}} = \gamma d_i^2 d_j/S_2$ would have to equal $\omega_j q_{\{j \rightarrow i\}} = \gamma d_j^2 d_i/S_2$, i.e. $d_i = d_j$. Since the sector dimensions differ,

$$q_{\{i \rightarrow j\}} = \gamma d_j \text{ cannot be detailed-balance for both } \pi \text{ and } \omega \text{ (verified: DB-}\pi \text{ holds, DB-}\omega \text{ fails).}$$

Quantum detailed balance, the GKLS detailed-balance adjoint, the jump-operator conditions, and the relative-entropy / gradient-flow structure are all defined *relative to a fixed faithful invariant state*. So π and ω require **two** generators, not a reinterpretation of one. This no-go is the reason v1.0's hope ("one L_π explains both") fails, and it sets up the correct bridge.

§U2. Trace and observer state are different functionals [PROVEN]. With the normalised matrix trace $\tau(x) = (1/D)\text{Tr}(x)$, the sector projectors carry

$$\tau(P_i) = d_i/D = (3, 2, 6)/11 = \pi,$$

the Part-I stationary (this is the corrected Gate-2 statement, §12). The observer weight is the **state** $\omega(x) = \tau(h x)$ defined by the modular density

$$h = \oplus_i (D d_i/S_2) I_{\{d_i\}}, \quad S_2 := \sum_i d_i^2 = 49, \quad \tau(h) = 1, \quad \omega(P_i) = \tau(h P_i) = d_i^2/S_2 = (9, 4, 36)/49.$$

The modular flow of ω is $\sigma_t^\omega(x) = h^{\{it\}} x h^{\{-it\}}$, so on a sector block $C_{\{ij\}} = P_i C P_j$ it acts as $\sigma_t^\omega(C_{\{ij\}}) = (h_i/h_j)^{\{it\}} C_{\{ij\}} = (d_i/d_j)^{\{it\}} C_{\{ij\}}$ — **exactly the ZS-A23 / ZS-F19 modular ratio**, now recovered with no trace/state confusion. (Theorem A24.III-0.)

§U3. The observer-balanced generator L_ω [PROVEN]. Define microscopic jumps tilted by the modular half-density:

$$L^\omega_{\{j\beta \leftarrow i\alpha\}} = \sqrt{\gamma} (h_j/h_i)^{\{1/4\}} |j, \beta\rangle \langle i, \alpha| = \sqrt{\gamma} (d_j/d_i)^{\{1/4\}} |j, \beta\rangle \langle i, \alpha|,$$

with reverse $L^\omega_{\{i\alpha \leftarrow j\beta\}} = \sqrt{\gamma} (d_i/d_j)^{\{1/4\}} |i, \alpha\rangle \langle j, \beta|$, for adjacent sectors $(X-Z, Z-Y)$ and all microstate pairs. The coarse rate is $q^\omega_{\{i \rightarrow j\}} = \gamma d_j \sqrt{(d_j/d_i)}$ (each target sector j supplying d_j microstates), and at the coarse level

$$\omega_i q^\omega_{\{i \rightarrow j\}} = (\gamma/S_2) d_i^{\{3/2\}} d_j^{\{3/2\}} = \omega_j q^\omega_{\{j \rightarrow i\}} \text{ (symmetric } \Rightarrow \omega\text{-detailed-balance),}$$

so the coarse generator is reversible with stationary $\omega = (9, 4, 36)/49$ (Theorem A24.III-1).

Theorem A24.III-1b (full quantum detailed balance, PROVEN — v1.6). On the *full* register M_{11} (not merely the coarse 3×3), the GKLS generator L_ω with the jumps above is **both KMS-symmetric and GNS-symmetric** with respect to $\rho_\omega = \oplus_i (d_i/S_2) I_{\{d_i\}}$, is modular-covariant, and fixes ρ_ω :

$$(i) L_\omega^*(\rho_\omega) = 0;$$

$$(ii) [L_\omega, \sigma_t^\omega] = 0, \text{ since each jump is a modular eigenoperator, } \sigma_t^\omega(|j, \beta\rangle \langle i, \alpha|) = (d_j/d_i)^{\{it\}} |j, \beta\rangle \langle i, \alpha| \text{ (eigenvalue from } h_j/h_i = d_j/d_i);$$

$$(iii) \text{ GNS symmetry: } \langle X, L_\omega(Y) \rangle_{\text{GNS}} = \langle L_\omega(X), Y \rangle_{\text{GNS}} \text{ with } \langle X, Y \rangle_{\text{GNS}} = \text{Tr}(\rho_\omega X^\dagger Y);$$

$$(iv) \text{ KMS symmetry: } \langle X, L_\omega(Y) \rangle_{\text{KMS}} = \langle L_\omega(X), Y \rangle_{\text{KMS}} \text{ with } \langle X, Y \rangle_{\text{KMS}} = \text{Tr}(\rho_\omega^{\{1/2\}} X^\dagger \rho_\omega^{\{1/2\}} Y).$$

Proof: the paired modular relation $\rho_\omega^{\{1/2\}} (L^\omega_{\{j \leftarrow i\}})^\dagger \rho_\omega^{\{-1/2\}} = L^\omega_{\{i \leftarrow j\}}$ holds for every jump, and the jumps are modular eigenoperators, so the Frigerio–Gorini–Kossakowski–Verri / Fagnola–Umanità criterion gives GNS symmetry; GNS symmetry plus (ii) gives KMS symmetry. The explicit matrix-unit computation on M_{11} verifies all four: the weighted-adjoint residuals $\|G_{\text{GNS}} L_\omega - L_\omega^* G_{\text{GNS}}\|$ and $\|G_{\text{KMS}} L_\omega - L_\omega^* G_{\text{KMS}}\|$ and $\|L_\omega^*(\rho_\omega)\|$ are $\sim 10^{-18}$, and $\|[L_\omega, \sigma_t^\omega]\| = 0$ exactly (zs_a24_v2_1_audit.py). This upgrades v1.5's coarse-only detailed balance to a genuine quantum detailed-balance theorem; GNS and KMS symmetry are distinct conditions (Fagnola–Umanità), and both hold. Its full Liouvillian spectrum closes by the §6 invariant-subspace method (escape rates $R_i = \gamma \sum_{\{j \sim i\}} d_j \sqrt{(d_j/d_i)}$; recorded as an optional extension, §20).

§U4. The modular interpolation family — full microscopic QMS [DERIVED]. The coarse rate interpolates the two generators by $q^{\{s\}}_{\{i \rightarrow j\}} = q^\omega_{\{i \rightarrow j\}} (h_j/h_i)^s$, $s \in [0, 1/2]$, with stationary measure $\mu^{\{s\}}_{\{i\}} \propto \pi_i h_i^{\{2s\}}$ (one line: $\mu^{\{s\}}_{\{i\}} q^{\{s\}}_{\{i \rightarrow j\}} = \pi_i q^\omega_{\{i \rightarrow j\}} h_i^s h_j^s$ is

symmetric in $i \leftrightarrow j$). To make "a family of QMS on M_{11} " literal, the **microscopic** generator is defined for every s by tilting each jump:

$$L^{\wedge}\{(s)\}_{j \leftarrow i \alpha} = \sqrt{\gamma} (h_j/h_i)^{s/2} |j, \beta\rangle\langle i, \alpha|, \quad \text{with invariant state } \rho_s = (\oplus_i h_i^{2s} I_{\{d_i\}}) / \sum_k d_k h_k^{2s}.$$

Theorem A24.III-2 (interpolation family, PROVEN). For every $s \in [0, 1/2]$, $L_{\{s\}}$ is a GKLS generator on M_{11} with $\rho_s^{\wedge}\{1/2\} (L^{\wedge}\{(s)\}_{j \leftarrow i})^\dagger \rho_s^{\wedge}\{-1/2\} = L^{\wedge}\{(s)\}_{i \leftarrow j}$ (KMS detailed balance), invariant ρ_s , and sector weight $\mu^{\wedge}\{(s)\}_i = \text{Tr}(P_i \rho_s) \propto d_i h_i^{2s} = \pi_i h_i^{2s}$. The endpoints are $L_{\{0\}} = L_{\pi} (\rho_0 = I_D/D, \mu = \pi \propto d_i)$ and $L_{\{1/2\}} = L_{\omega} (\rho_{1/2} = \rho_{\omega}, \mu = \omega \propto d_i^2)$. *Proof:* the tilt is a modular half-density (Doob-type) transform; the displayed pairing is the GNS/KMS detailed-balance condition; verified at $s = 0, 1/4, 1/2$ on the full M_{11} (stationarity, the pairing, and GNS/KMS symmetry, residuals $\sim 10^{-18}$; zs_a24_v2_1_audit.py). A single modular half-density interpolation thus carries the state-sampling dynamics to the operator-sampling dynamics through a continuum of genuine quantum Markov semigroups.

§U5. Central Theorem [DERIVED — finite-register].

Theorem A24.III (finite-register trace/state unification). The state sampling $\pi \propto d_i$ and the observer operator sampling $\omega \propto d_i^2$ are **not one stationary state** but the $s = 0$ and $s = 1/2$ endpoints of **one modular half-density interpolation** $L_{\{s\}}$, a family of **QMS on M_{11} that preserve the block-diagonal subalgebra $N = M_3 \oplus M_2 \oplus M_6$** (the cross-sector jumps $|j, \beta\rangle\langle i, \alpha|$ are off-diagonal elements of M_{11} , not of N ; restricting each $L_{\{s\}}$ to N gives the sector population process). With the single matrix trace τ on M_{11} , they are two reversible quantum dynamics joined by the modular half-density transform $h = \oplus_i (D d_i/S_2) I_{\{d_i\}}$: L_{ω} is obtained from L_{π} by tilting each jump by $(h_j/h_i)^{1/4}$, $\omega(\cdot) = \tau(h \cdot)$, and (Theorem A24.III-1b) L_{ω} is KMS- and GNS-symmetric and modular-covariant. *Proof:* §§U1–U4.

This is the complete unification **at the finite-register level**. Part I is the *dynamics* (how the register relaxes, stationary π); Part II is the *observer* (operator-space size-bias, ω); the **bridge** is the modular density h that both defines the A23 modular flow *and* maps L_{π} to L_{ω} . The conflation corrected in §12 (treating ω as the trace) was precisely what had hidden this: once the trace is π and ω is the modular state $\tau(h \cdot)$, the two Parts are visibly one family.

§U6. What is NOT yet unified — the continuous-core dynamical lift [OPEN]. PART III lives on the finite register M_{11} . Part II's actual object is the continuous-core corner $M_{\text{obs}} = e C_{\omega} e$, $C_{\omega} = \mathcal{A}^{\text{III}} \rtimes_{\sigma} \mathbb{R}$. §11 supplies a conditional expectation $E : M_{\text{obs}} \rightarrow \mathfrak{U}(M_{11})$ intertwining the *population* generator ($E \circ \mathcal{L} = \mathcal{L}_{\text{pop}} \circ E$). What is **not** constructed is the dynamical lift of the *bridge*: a generator $\mathcal{L}_s$ on M_{obs} with

$$E \circ \mathcal{L}_s = L_s \circ E \quad (\text{intertwining}), \text{ plus modular-clock covariance, complete positivity, core-trace preservation, and extension to the } \mathbb{Z}_2\text{-graded core.}$$

PART III thus proves the **finite-register modular unification** and supplies the **candidate finite-dimensional dynamics to be lifted**; the continuous-core dynamical unification is **OPEN** (F-A24.9). This is consistent with the Book/ZS-F23, which keep Condition C (the identification of a specific gravitational subregion algebra with the finite register) open under current tools. **[Honest scope:**

"complete Part I ↔ Part II unification" would be an over-claim; the proven statement is the finite-register unification plus a conditional lift.]

§16. Cross-Version Consistency Audit

Table 2. Cross-version dependency audit (the buildup check of protocol §3.2).

Upstream result	Status	v1.5's effect	Safe?
ZS-A19/M6 $\kappa^2 = A/Q = 35/4807$	PROVEN	the single generator coupling; unchanged	✓
ZS-M1 i-tetration z^*	PROVEN	not consumed; spectrum anchor only	✓
A23.5a/A23.14a $\pi = (3, 2, 6)/11$	PROVEN	Part I §3: stationary of the general generator	✓
ZS-Q7 cubic $\{0, -2A/Q, -A\}$	PROVEN	Part I §6: the population block of the 121-spectrum	✓
A23.MC $\omega = (9, 4, 36)/49$, $\sigma_t^* \omega = (d_i/d_j)^{\{it\}}$	PROVEN	§12/PART III: $\omega = (9,4,36)/49$ is the state $\tau(h\cdot)$; the corner trace is $\pi = (3,2,6)/11$	✓
Takesaki dual weight (dual weight = trace $\times$ density)	IMPORTED-PROVEN	§12: corrects v1.4 — the dual weight is $\tau(h\cdot)$, so ω is the dual-weight state , not the trace	✓
A24 §6 invariant-subspace decomposition	PROVEN (this work)	general exact spectrum A24.I-4G; 121×121 becomes a regression test	✓
A24 PART III modular half-density bridge	DERIVED (finite-register)	$L_\pi(\pi)$ and $L_\omega(\omega)$ unified as $s = 0, \frac{1}{2}$ endpoints on M_{11} ; continuous-core lift OPEN (F-A24.9)	✓
A24 §U3 full quantum detailed balance of L_ω	PROVEN (this work)	KMS- and GNS-symmetric on M_{11} , $[L_\omega, \sigma_t^* \omega] = 0$, $L_\omega^*(\rho_\omega) = 0$ (matrix-unit, $\sim 10^{-18}$)	✓
Fagnola–Umanità GNS/KMS detailed-balance structure	IMPORTED-PROVEN	§U3: criterion for GNS $\Rightarrow$ KMS symmetry of L_ω ; the bridge machinery is standard (positioning)	✓
A23.15 modular detailed balance $-\Delta K$	PROVEN	Part I §4; the KMS link to the modular clock	✓
ZS-F19 $\Delta K_{\{X \rightarrow Y\}} = -\ln 2$, lattice = UV regulator	DERIVED	§10: II_∞ core regulator; §12: offset matching	✓
ZS-F23 $\mathbb{Z}_2$ seam grading (J, J_Z)	PROVEN	§10: the trace-preserving β	✓
ZS-F2 §11.4 $\Omega_\Lambda = 83/121$, Thm 11.7	DERIVED	§13: partition (6,32,83) = (F(cube), F(TI), rem.) DERIVED ; Gate 3 $\rightarrow$ DERIVED-COND.	✓
CLPW de Sitter Type II ₁ , max-entropy = empty dS	IMPORTED-PROVEN	§10–§12: the continuous core and offset	✓
ZS-F1 Z-Anchor vortex (global; §6.4 Λ)	PROVEN	§14: equal-edge doubling DERIVED; vortex route closed $\rightarrow$ per-edge $= \pi/A$ (ZS-M3); geometric Λ	✓
A23 G12b-exponent $1/\kappa^2 = Q/A$	PROVEN	§14: RETRACTED as the per-edge action (numerology; no corpus instanton $= 1/\kappa^2$); stays the LOCKED inverse-coupling	✓ (corrected)

Upstream result	Status	v1.5's effect	Safe?
ZS-M3 §6 $t_{\text{flip}} = \pi/A$, $T_{\text{micro}} = 2\pi/A$, $S_{\text{tunnel}} = 5\pi/A$	DERIVED	§14: per-edge = π/A (seam half-event), $S_{\{X \rightarrow Y\}} = 2\pi/A = T_{\text{micro}}$, factor 2 = two half-events	✓
ZS-F0 §8.5 Wilson loop = BV–BFV cobordism invariant	DERIVED	§13: Z-sector partition function is topological $\Rightarrow$ holography combinatorial, not metric	✓
v1.0 A24.1a ($M_{11} \rtimes \mathbb{R}$ Type II)	RETRACTED	§9: finite route CLOSED-NEGATIVE; replaced by §10–§11	✓ (corrected)

Result. No version conflict. v1.3 *consumes* the locked anchors with zero new fitted parameters; it consumes ZS-M3 §6 (the seam-flip per-edge action), ZS-A23.MC (the operator-sampling weight), and ZS-F0 §8.5 (the cobordism/combinatorial holography), and it retracts only A24's own over-claims (the v1.0 finite route and the A23 per-edge = $1/\kappa^2$ guess). Nothing upstream is disturbed. **[DERIVED.]**

§17. Consolidated Gate Ledger

Table 3. Consolidated gate ledger (v2.1).

Gate	Calculation	Status (v1.5)
G-I.general (A24.I-1)	$q_{\{ij\}} = \Gamma_0 \kappa^2 a_{\{ij\}} d_j$ on any connected graph; $\pi = d/\Sigma d$	PROVEN.
G-I.Htheorem (A24.I-3)	exact H-theorem; $\mathbf{Q}/2$; reverse rates 2:3:6:2	PROVEN.
G-I.uniqueness (A24.I-uniq)	one scalar per edge (Schur, PROVEN) + global edge-homogeneity $\Rightarrow$ unique	DERIVED (the full uniqueness rests on the single- κ^2 action-level homogeneity input; the per-edge Schur step alone is PROVEN).
G-I.Liouvillian (A24.I-4)	full 121×121 spectrum $\{0, -2A/Q, -A/2, -9A/Q, -A\}$, mult (1,80,36,3,1)	PROVEN (NEW).
G-I.spectrum (A24.I-4G)	general exact $\text{Spec}(\mathbf{L})$ for any (a,b,c); Z-Spin $\{0, -2A/Q, -A/2, -9A/Q, -A\}$	PROVEN (v1.5, representation-theoretic; 121×121 = regression).
G-I.stability (A24.I-5)	reciprocal: π exact, gap ($1 \pm \epsilon$); directional 3-state: exact π , $g_{\pm}$, TV, forward-only gap; generic: numerical	PROVEN (§7.1–7.2) / VERIFIED-NUMERICAL (§7.3).
G-II.finite-route (A24.II-0)	$M_{11} \rtimes_{\sigma} \mathbb{R}$ is Type I (inner action)	CLOSED-NEGATIVE; v1.0 claim RETRACTED.
G-II.core (A24.II-1)	$\mathbb{R} \rtimes_{\sigma} \mathbb{R} = \Pi_{\infty}$ continuous core	IMPORTED-PROVEN (Witten/CLPW/Takesaki).
G-II.corner (A24.II-2)	$e\mathbf{C}_{\omega} e = \Pi_1$; $\iota: M_{11} \hookrightarrow M_{\text{obs}}$	DERIVED-CONDITIONAL on the embedding.
G2(i) observer trace/state (A24.II-3)	corner trace = matrix trace = $\pi = (3,2,6)/11$; observer weight $\omega = \tau(h \cdot) = (9,4,36)/49$	DERIVED (corrected v1.5: trace = π , ω = modular state via h ; restores A23 consistency).

Gate	Calculation	Status (v1.5)
G-III unification (A24.III)	$L_\pi(\pi)$ and $L_\omega(\omega)$ joined by modular half-density h on M_{11} ; one family $L_{\{s\}}$, $s \in [0, \frac{1}{2}]$	DERIVED (finite-register) (no-go PROVEN, L_ω PROVEN, interpolation DERIVED).
G-III.DB full quantum DB (A24.III-1b)	L_ω KMS- and GNS-symmetric on M_{11} ; $[L_\omega, \sigma_{t^\omega}] = 0$; $L_\omega^*(\rho_\omega) = 0$	PROVEN (matrix-unit, v1.6).
G-III.lift continuous-core (F-A24.9)	$\exists \mathcal{L}_s$ on M_{obs} with $E \circ \mathcal{L}_s = L_s \circ E +$ modular covariance + CP + core-trace	OPEN (the single honest gap; candidate finite dynamics supplied).
G2(ii) CLPW offset (A24.II-4)	$S_0 = A_{\text{dS}}/4G$; residual = B3	IMPORTED-MATCHING / DERIVED-CONDITIONAL.
G3 face partition (§13)	$(6, 32, 83) = (F(\text{cube}), F(\text{TI}), \text{rem.})$, ZS-F2 §11.4	DERIVED ; realised as Π_1 trace-1/121 projections.
G3 face weight (§13)	one quantum per face = combinatorial (not metric)	combinatorial nature DERIVED (ZS-F0 §8.5 cobordism); residual = F0's deferred explicit state-sum.
G4 factor 2 (A24.II-5)	$S_{\{X \rightarrow Y\}} = 2 S_{\text{vortex}}$ (equal edges)	DERIVED (topological Z-Anchor).
G4 per-edge (A24.II-7)	$1/\kappa^2$ RETRACTED; per-edge = π/A (ZS-M3 $\mathbb{Z}_2$ half-event)	per-edge = π/A DERIVED-CONDITIONAL ; $S_{\{X \rightarrow Y\}} = 2\pi/A = T_{\text{micro}}$; $1/\kappa^2 \leftrightarrow \Lambda$ numerology retracted.
G-B3 (§15.1)	absolute scale $\bar{H}/\bar{M}_P$	OPEN — corpus-wide debt.
G-C5 (§15.2)	clock DOF $\propto \mathbb{R}$ as Quintom field (i)	HYPOTHESIS-weak — B1 reflection-positivity pending.

§18. Falsification Gates

Multilayer falsification (protocol §3.5): mathematical/theoretical, simulation/consistency, observational/external.

- **F-A24.1 (mathematical — now correctly read).** "If σ^ω is inner, the finite crossed product is Type I and a finite-route Type II claim fails." $\sigma^\omega = \text{Ad}(\rho_\omega^{\{it\}})$ is inner (Skolem–Noether); F-A24.1 is **TRIGGERED**, and the finite-route claim is retracted (§9). The genuine route (§10) starts from the Type III₁ algebra whose modular flow is outer, and is not subject to this gate.
- **F-A24.2 (mathematical).** If the full Liouvillian spectrum is **not** $\{0, -2A/Q, -A/2, -9A/Q, -A\}$ with multiplicities (1, 80, 36, 3, 1), Theorem A24.I-4G fails. *Check:* the representation-theoretic theorem matches the 121×121 build and the (2,3,4) control (audit). [Cleared.]
- **F-A24.3 (mathematical).** If $\pi = d/\Sigma d$ is **not** stationary/reversible for a connected graph, Theorem A24.I-1 fails. *Check:* three test graphs pass (audit I.1). [Cleared.]
- **F-A24.4 (instanton — the priority gate, resolved in v1.3).** If the per-edge action differs from the claimed value, the claim fails. *Result (v1.3):* the per-edge = $1/\kappa^2$ claim is **RETRACTED** — no corpus instanton has action $1/\kappa^2 = Q/A \approx 137.34$ (the corpus actions are $t_{\text{flip}} = \pi/A$, $T_{\text{micro}} =$

$2\pi/A$, $S_{\text{tunnel}} = 5\pi/A$, $S_{\text{cl}} = 35\pi/3$; $Q/\pi \approx 3.5$ is not a flip count), and the $1/\kappa^2 \leftrightarrow \Lambda$ motivation was numerology. Since the A24 mediation crosses the $\mathbb{Z}_2$ seam (ZS-F23), the corrected per-edge action is the **DERIVED seam-flip $t_{\text{flip}} = \pi/A$** (ZS-M3 §6), with $S_{\{X \rightarrow Y\}} = 2\pi/A = T_{\text{micro}}$ and **factor 2 = two half-events** (DERIVED). [Fired against $1/\kappa^2$; the per-edge value is corrected to the corpus seam-flip π/A .]

- **F-A24.5 (consistency — observer trace/state, corrected v1.5).** If the corner trace were not the matrix trace $\pi = (3, 2, 6)/11$, or the observer state $\omega(\cdot) = \tau(h \cdot)$ not equal to $(9, 4, 36)/49$, Gate 2 would fail. *Status (v1.5): DERIVED* — $\tau(P_i) = d_i/D = (3, 2, 6)/11$ is the matrix trace, and $\omega(P_i) = \tau(h P_i) = d_i^2/\Sigma d_j^2 = (9, 4, 36)/49$ with $h_i = D d_i/\Sigma d_j^2$ (both verified). v1.4's identification of the corner *trace* with ω is **corrected**: the dual weight is the trace times h , so ω is the dual-weight *state*. This restores consistency with ZS-A23 (trace weights = $(3, 2, 6)/11$; $\omega_{\text{reg}} = (9, 4, 36)/49$) and enables the PART III unification.
- **F-A24.6 (observational — Part I).** A small 3-2-6 open-system simulator measuring $p(\infty) \neq (3, 2, 6)/11$, or $\lambda_{\text{fast}}/\lambda_{\text{slow}} \neq 11/2$, or cross-sector coherence decay $\neq A/2$, falsifies the Part I dynamics. [TESTABLE now.]
- **F-A24.7 (unification — PART III).** If a single generator could be detailed-balance for both π and ω (contradicting the no-go), or if the tilted generator L_ω with $q^\omega_{i \rightarrow j} = \gamma d_j \sqrt{(d_j/d_i)}$ did not have stationary $\omega = (9, 4, 36)/49$, or if the interpolation $\mu^\wedge\{s\} \propto \pi_i h_i^{2s}$ did not run from π ($s = 0$) to ω ($s = 1/2$), the modular half-density bridge fails. *Check*: the no-go, L_ω detailed balance, and the interpolation endpoints all verify (audit). [Cleared.]
- **F-A24.8 (observational — C5).** A confirmed healthy phantom-divide crossing (DESI + global model comparison) activates the clock-DOF C5 candidate (§15.2), which must then provide the healthy crossing or be excluded by B1.
- **F-A24.9 (continuous-core lift — OPEN).** The finite-register unification (PART III) lifts to the continuous core only if there exists a generator $\mathcal{L}_s$ on $M_{\text{obs}} = eC_\omega e$ with $E \circ \mathcal{L}_s = L_s \circ E$ (intertwining the §11 conditional expectation), modular-clock covariance, complete positivity, core-trace preservation, and a $\mathbb{Z}_2$ -graded extension. No such L_s is constructed here. *Status: OPEN* — the single honest gap of A24; its closure would promote "finite-register unification" to "Part I $\leftrightarrow$ Part II dynamical unification." Consistent with the Book/ZS-F23 keeping Condition C open.

§19. Conclusion

ZS-A23 reduced the corpus's dynamical core to one dimension-weighted mediator generator. v1.0 of A24 attempted to lift it to a de Sitter observer algebra but mis-identified a finite inner crossed product as Type II. The genuine Type III₁ $\rightarrow$ II $_\infty$ continuous core corrects that, and the present work develops the programme in three Parts — the dimension-weighted dynamics (I), the honest continuous-core lift (II), and the finite-register modular bridge (III).

Part I is a self-contained, PROVEN theory of dimension-weighted mediator semigroups: a general connected-graph stationary law, modular detailed balance, an exact H-theorem, a unique sector-covariant GKLS realisation, an **exact general Liouvillian spectrum** (Theorem A24.I-4G, any (a, b, c) ; for Z-Spin $\{0, -2A/Q, -A/2, -9A/Q, -A\}$ with the cross-sector coherences decaying at the clean rate $A/2$), and **two PROVEN perturbation theorems** (a reciprocal Dirichlet-form gap bound and an exact directional 3-state analysis with a forward-only first-order gap shift), generic microscopic errors staying numerical.

Every relaxation and decoherence rate is a rational multiple of $\mathbf{A}$, with zero new parameters; the whole of Part I is testable on a small open-system simulator.

Part II rebuilds the lift on the correct algebraic footing. The finite route is **retracted** (an inner modular flow cannot generate a Type II factor); the genuine continuous core is the Type $\text{III}_1 \rightarrow \text{II}_\infty$ crossed product $\mathcal{A}^{\text{III}} \rtimes_{\sigma} \mathbb{R}$ (IMPORTED-PROVEN), with a finite II_1 corner $e\mathbf{C}_{\omega}e$ and a register embedding $\iota : M_{11} \hookrightarrow e\mathbf{C}_{\omega}e$ (DERIVED-CONDITIONAL). The register embeds, $\iota : M_{11} \hookrightarrow e\mathbf{C}_{\omega}e$ (DERIVED-CONDITIONAL on the embedding's existence and modular-density compatibility; the corner trace is automatically the matrix trace $\pi = (3, 2, 6)/11$). The observer weight $\omega = (9, 4, 36)/49$ is recovered **not** as a trace but as the modular *state* $\tau(h \cdot)$ (v1.5/§12), and the CLPW offset $\mathbf{A}_{\text{dS}}/4G$ under an empty-de-Sitter reference (IMPORTED-MATCHING). The face-embedding target is realised in II_1 continuous trace dimension ($\tau(q_a) = 1/121$, coarse trace $(6, 32, 83)/121$), well-posed where the v1.0 rank statement was impossible; **v1.2 grounds the partition in ZS-F2 §11.4** — $(6, 32, 83) = (\text{F(cube)}, \text{F(truncated icosahedron)}, \text{remainder})$ — and **v1.3 resolves the equal-weight as combinatorial** (not metric) holography, since ZS-F0 §8.5 makes the Z-sector partition function a BV–BFV cobordism invariant (Gate 3: DERIVED-CONDITIONAL, residual = F0's deferred explicit state-sum). The Gate-2 observer weight $(9, 4, 36)/49$ is **PROVEN** as the operator-sampling weight (ZS-A23.MC), but **v1.5 corrects its operator-algebra identity**: the corner **trace** is the matrix trace $\pi = (3, 2, 6)/11$ (the Part-I stationary), while $\omega = (9, 4, 36)/49$ is the **state** $\tau(h \cdot)$ given by the modular density h (v1.4 had conflated this state with the trace, contradicting its own audit and ZS-A23). The surviving residuals are explicitly upstream (ZS-F0's deferred state-sum, CLPW's reference state), corpus-wide (B3), or registered (C5). The two-edge instanton **factor 2 is DERIVED**, and **v1.3 corrects the per-edge value**: $1/\kappa^2$ is **RETRACTED** (no corpus instanton equals it; the $1/\kappa^2 \leftrightarrow \Lambda$ match was numerology), and the corpus-grounded per-edge action is the $\mathbb{Z}_2$ **seam-flip $t_{\text{flip}} = \pi/\mathbf{A}$** (ZS-M3 §6, DERIVED), with $S_{\{X \rightarrow Y\}} = 2\pi/\mathbf{A} = T_{\text{micro}}$ and the factor 2 = two half-events. Λ remains geometric (ZS-F1 §6.4), decoupled from the seam action.

Part III is the unification the two-Part split had gestured at — **at the finite-register level**. A no-go shows a single generator cannot detailed-balance both the state weight $\pi \propto d_i$ and the operator weight $\omega \propto d_i^2$. The correct picture (forced by the §12 trace/state correction) is **two** reversible dynamics on the *same* register algebra M_{11} with the *same* trace π : the dimension-weighted $\mathbf{L}_{\pi}$ (stationary π) and the observer-balanced $\mathbf{L}_{\omega}$ (stationary ω , jumps tilted by $(h_j/h_i)^{1/4}$), joined by the **modular half-density** h and embedded in one interpolation family $\mathbf{L}_{\{s\}}$, $s \in [0, 1/2]$. v1.6 proves $\mathbf{L}_{\omega}$ is a genuine quantum-detailed-balance generator (KMS- and GNS-symmetric on M_{11} , modular-covariant). The central theorem A24.III states it plainly: *finite-register state sampling and observer operator sampling are not one stationary state but two reversible quantum dynamics joined by a modular half-density transform*. This unifies Part I (the dynamics) and Part II (the observer) **on the finite register**, and supplies the candidate dynamics to be lifted; the **continuous-core dynamical lift** — a generator $\mathcal{L}_s$ on $M_{\text{obs}} = e\mathbf{C}_{\omega}e$ intertwining the conditional expectation with modular covariance and complete positivity — remains the single honest **OPEN** item (F-A24.9).

Relation to standard constructions. The individual tools used here — the modular (Doob-type) half-density tilt of a generator, KMS- and GNS-symmetric quantum Markov semigroups, and Dirichlet-form / Rayleigh–Ritz spectral-gap bounds — are **standard** (Fagnola–Umanità; the quantum-Doob / tilted-Lindblad literature). A24 does **not** claim them as new mathematics. Its contribution is the *exact package* assembled on one specific object: the dimension-weighted 3-2-6 register, where the dimension multiplicity fixes the exact Liouvillian spectrum (A24.I-4G), the two

stationary samplings ($\pi \propto d$, $\omega \propto d^2$) are the modular-bridge endpoints, and the whole is a concrete small-simulator proposal. The "first exact spectrum of the isotropic complete-bipartite Lindbladian" framing is stated as a clean reusable proposition pending a fuller literature search, not as a major-problem resolution.

Net. The semigroup is a complete open-quantum-systems theory with an exact general spectrum, two PROVEN perturbation theorems, and a full quantum-detailed-balance theorem for the observer generator; the continuous-core lift is correctly founded; **Parts I and II are unified by the modular half-density bridge at the finite-register level**, with the continuous-core dynamical lift the single honest OPEN item (F-A24.9). The cosmological-constant problem is localised to the same named, falsifiable items — the Gate-2(ii) reference state, the B3 absolute scale, and the per-edge instanton normalisation. The corpus standard is maintained: the central trace-vs-state conflation carried from v1.0 is corrected, the central claim is bounded to what is proved, anti-numerology enforced, every over-claim retracted or made conditional. Zero free parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ **LOCKED**.

§20. Remaining Programs

As of v2.0: the finite-register mathematics is final; the single honest OPEN item is the continuous-core dynamical lift. The remaining items are upstream, corpus-wide, registered, or optional extensions.

3. **(continuous-core dynamical lift — the single honest OPEN, F-A24.9)** Construct a generator $\mathcal{L}_s$ on $M_{\text{obs}} = eC_{\omega}e$ with $E \circ \mathcal{L}_s = \mathcal{L}_s \circ E$ (intertwining the §11 conditional expectation), modular-clock covariance, complete positivity, core-trace preservation, and a $\mathbb{Z}_2$ -graded extension. PART III supplies the candidate finite-dimensional dynamics; this lift would promote the finite-register unification to a genuine Part I $\leftrightarrow$ Part II dynamical unification. Consistent with the Book's open Condition C.
4. **(embedding — Gates 2, 3: A24-internal part CLOSED)** Gate 2 is DERIVED with the corrected trace/state reading (§12); Gate 3's partition and combinatorial nature are DERIVED. The single surviving embedding item is **upstream, not A24's**: the *explicit* ZS-F0 face-weight state-sum (a Turaev–Viro / Barrett–Westbury realisation forcing the equal 1/121 weight) is ZS-F0's own deferred categorical formalisation (F0 §166). A24 has supplied the local monoidal-functor argument; the categorical polish is ZS-F0's.
5. **(per-edge — Gate 4: scale DERIVED)** The per-edge action is the ZS-M3 seam-flip at the π/A scale (same-seam lemma + ZS-M3 §8, DERIVED), with per-edge = $t_{\text{flip}} = \pi/A$ and $S_{\{X \rightarrow Y\}} = T_{\text{micro}} = 2\pi/A$. The only residual is the **non-load-bearing** leg-count (π/A vs $\pi/2A$); the cosmological factor 2 and the scale are DERIVED, and Λ is geometric (ZS-F1 §6.4), decoupled.
6. **(Gate 2(ii) — upstream)** The CLPW entropy offset $S_0 = A_{\text{dS}}/4G$ is IMPORTED-MATCHING from CLPW's empty-de-Sitter reference state.
7. **(B3)** The action-level absolute IR scale $\bar{H}/\bar{M}_P$ (the corpus-wide debt; not local to A24).
8. **(C5)** A reflection-positivity proof (or B1 exclusion) for the clock-sector EFT.
9. **(L_{ω} full spectrum — optional extension)** The observer-balanced L_{ω} (PART III) has an exact spectrum by the §6 invariant-subspace method (sector-internal traceless modes at $-R_i$, multiplicity

$d_{-i}-1$; cross-sector coherences at $-(R_{-i}+R_{-j})/2$, multiplicity $2d_{-i}d_{-j}$; and the 3-state identity block giving 0, $-g_{-}\omega$, $-g_{+}\omega$ from the §7.2 formula), with $R_{-i} = \gamma \sum_{j \sim i} d_j \sqrt{(d_j/d_{-i})}$. Writing it out gives a second full Liouvillian theorem; not required for the unification.

10. **(full graded Liouvillian)** Extend to the spin-graded core $C_{\omega} \rtimes \beta$ and the continuous-clock sector, with the modular-flow phases $(d_{-i}/d_{-j})^{\{it\}}$ as the conservative companion to the §6 dissipative spectrum.

Appendix A — General Proof of the Stationary Law (Theorem A24.I-1)

For a connected undirected graph with adjacency $a_{\{ij\}} = a_{\{ji\}}$ and node multiplicities d_i , set $q_{\{i \rightarrow j\}} = \Gamma_0 \kappa^2 a_{\{ij\}} d_j$ ($i \neq j$), $q_{\{i \rightarrow i\}} = -\sum_{k \neq i} q_{\{i \rightarrow k\}}$. Detailed balance with $\pi_{-i} = d_{-i}/D$ ($D = \sum_k d_k$): $\pi_{-i} q_{\{i \rightarrow j\}} = (d_{-i}/D) \Gamma_0 \kappa^2 a_{\{ij\}} d_j = (\Gamma_0 \kappa^2/D) a_{\{ij\}} d_i d_j$, which is symmetric in i, j ($a_{\{ij\}} = a_{\{ji\}}$); hence $\pi_{-i} q_{\{i \rightarrow j\}} = \pi_{-j} q_{\{j \rightarrow i\}}$. Summing over j : $\sum_j \pi_{-i} q_{\{i \rightarrow j\}} = \sum_j \pi_{-j} q_{\{j \rightarrow i\}}$, i.e. $(\pi Q)_{-i} = 0$, so π is stationary. Connectedness $\Rightarrow$ the chain is irreducible $\Rightarrow \pi$ is unique. Reversibility $\Rightarrow \mathcal{L}$ is self-adjoint in the π -weighted inner product, so the spectrum is real and the relaxation is monotone (Theorem A24.I-3).

Appendix B — The Full 121×121 Liouvillian (Theorem A24.I-4)

Hilbert space $\mathbf{H} = \mathbb{C}^3 \oplus \mathbb{C}^2 \oplus \mathbb{C}^6$ ($X = \{0,1,2\}$, $Z = \{3,4\}$, $Y = \{5,\dots,10\}$). Jump operators $L_{\{(j) \leftarrow (i)\}} = \sqrt{(\Gamma_0 \kappa^2) |j\rangle \langle i|}$ for every ordered adjacent pair on the path $X \leftrightarrow Z \leftrightarrow Y$ ($X \leftrightarrow Z$ and $Z \leftrightarrow Y$, both directions; no $X \leftrightarrow Y$): $6 + 6$ ($X \leftrightarrow Z$) + $12 + 12$ ($Z \leftrightarrow Y$) = 36 operators. The Liouvillian on $\mathcal{L}(\mathbf{H}) \cong \mathbb{C}^{121}$ is, in column-stacking convention,

$$\mathcal{L} = \sum_k [\bar{L}_k \otimes L_k - \frac{1}{2}(I \otimes L_k^\dagger L_k + (L_k^\dagger L_k)^T \otimes I)].$$

By the analytic Theorem A24.I-4G (§6, invariant-subspace decomposition) the spectrum is, for Z-Spin, $\text{Spec}(\mathcal{L}) = \{0, -2\mathbf{A}/\mathbf{Q}, -\mathbf{A}/2, -9\mathbf{A}/\mathbf{Q}, -\mathbf{A}\}$ with multiplicities (1, 80, 36, 3, 1), $\Sigma = 121$. As an **independent reproduction** (regression test, not the derivation), numerical diagonalisation of the 121×121 superoperator at $\Gamma_0 \kappa^2 = 1$ returns exactly $\{0, -2, -11/2, -9, -11\} \times \kappa^2$ with the same multiplicities, all eigenvalues real ($\max |Im| < 10^{-9}$). The 36-fold $-\mathbf{A}/2$ block is the cross-sector coherence space ($X \leftrightarrow Z$: 6+6; $Z \leftrightarrow Y$: 12+12 = 36). Reproduced by `zs_a24_v2_1_audit.py` (the analytic theorem and the numerical spectrum agree for (3, 2, 6) and the (2, 3, 4) control).

Appendix C — The Bogomolnyi Per-Edge Action (Computation A24.II-6)

v1.2 — the Z-Anchor is a global vortex. ZS-F1 §7 keeps the Goldstone massless and physical ($\Delta N_{\text{eff}} = 0$), so the $U(1)$ is global and the energy is **log-divergent**, $E \approx \pi v^2 \ln(L/\xi)$ — the BPS-finite tension below is the *gauged counterfactual*, recorded to show the per-edge value even in the most favourable (BPS) reading. Critically-coupled abelian Higgs model ($\lambda = e^2/2$), Euclidean action $S = \int d^2x [\frac{1}{4}F^2 + |D\phi|^2 + (\lambda/4)(|\phi|^2 - v^2)^2]$. Bogomolnyi completion: $S = \frac{1}{2} \int [F_{12} \mp e(|\phi|^2 - v^2)]^2 + |D_1\phi \mp iD_2\phi|^2 \pm e v^2 \int F_{12}$, with bound $S \geq e v^2 \int |F_{12}| = e v^2 |\Phi_B|$, $\Phi_B = 2\pi n/e$ (flux quantisation). Saturation: $S_{\text{vortex}} = 2\pi v^2 |n|$. With $v = 1$ ($\bar{M}_P$ units), $n = 1$: $S = 2\pi \approx 6.283$. Target $1/\kappa^2 = \mathbf{Q}/\mathbf{A} = 4807/35 \approx 137.343$. Condition for $S_{\text{vortex}} = 1/\kappa^2$: $v^2 |n| = 1/(2\pi\kappa^2) = \mathbf{Q}/(2\pi\mathbf{A}) = 4807/(70\pi) \approx 21.859$ — i.e. $n \approx 22$ ($v = 1$), excluded by the minimal stable winding $n = 1$ (ZS-F1 §5). Both readings (global log-divergence; gauged 2π) miss $1/\kappa^2$, so the vortex

per-edge route is closed. **v1.3:** the corpus instanton ledger ($t_{\text{flip}} = \pi/A$, $T_{\text{micro}} = 2\pi/A$, $S_{\text{tunnel}} = 5\pi/A$, $S_{\text{cl}} = 35\pi/3$) contains no action equal to $1/\kappa^2$ ($Q/\pi \approx 3.5$ is not a flip count), so per-edge = $1/\kappa^2$ is **RETRACTED**; since the A24 mediation crosses the $\mathbb{Z}_2$ seam (ZS-F23), the corrected per-edge action is the ZS-M3 §6 half-event **$t_{\text{flip}} = \pi/A$** , with $S_{\{X \rightarrow Y\}} = 2\pi/A = T_{\text{micro}}$. Reproduced by `zs_a24_v2_1_audit.py` checks F4.b–c, the global-vortex check, and the corpus-instanton-ledger check.

Appendix D — Anti-Numerology Pre-Registration

MC #1 — the face triple (6, 32, 83). Uniform random ordered 3-partition of 121 into positive parts; $P(\text{unordered set} = \{6, 32, 83\}) = 6/C(120, 2) = 6/7140 \approx 8.40 \times 10^{-4}$ (Monte-Carlo 2×10^5 : 0.00090). $P \ll 5\%$, so the Gate-3 target is **non-generic**; a future Φ_{face} derivation landing on it is meaningful, not a fit. (A24 does not derive it.)

MC #2 — the per-edge value $1/\kappa^2 = Q/A \approx 137.343$. The coincidence with $1/\alpha_{\text{em}} \approx 137$ is reported as an **OBSERVATION** and used in no derivation. The value is the LOCKED inverse-coupling, not a fit. **v1.3 note:** $1/\kappa^2$ is *not* the per-edge instanton action — that identification (and the $e^{\{-2/\kappa^2\}} \approx \Lambda$ motivation behind it) is **RETRACTED** as numerology (§14.2); the corpus per-edge action is π/A (ZS-M3 §6). $1/\kappa^2$ remains only the inverse-coupling.

Cross-version safety. z^* (ZS-M1), $\kappa^2 = 35/4807$ (ZS-A19/M6), $\pi = (3, 2, 6)/11$ (A23.5a), $\omega = (9, 4, 36)/49$ (A23.MC), the population spectrum $\{0, -2A/Q, -A\}$ (ZS-Q7) are unchanged and consumed; zero new fitted parameters.

Acknowledgements and Code Availability

This v2.1 is the consolidated public release; the full development is recorded in Appendix E and the Version History. The Gate-2 operator-algebra identity is correct — the Π_1 -corner *trace* is the matrix trace $\pi = (3, 2, 6)/11$, while the observer weight $\omega = (9, 4, 36)/49$ is the *state* $\tau(h \cdot)$ given by the modular density h — and PART III builds the **modular half-density bridge**. The central claim is **bounded to the finite-register level** (the continuous-core dynamical lift is the single honest OPEN item, F-A24.9), **proves the full quantum detailed-balance theorem** (Theorem A24.III-1b: L_{ω} is KMS- and GNS-symmetric on M_{11} , modular-covariant, ρ_{ω} -stationary, by explicit matrix-unit computation), and fixes the residual textual errors. No new physics is added. The continuous-core construction follows Takesaki's continuous decomposition and the gravitational crossed-product programme of Witten and Chandrasekaran–Longo–Penington–Witten; the trace/state split and modular density follow Tomita–Takesaki theory; the GNS/KMS detailed-balance structure follows Fagnola–Umanità and the tilted-Lindblad / quantum-Doob literature (standard machinery — A24's contribution is the exact package on the specific 3-2-6 register); the spectrum theorem follows the Lindblad/representation-theory of the GKLS generator; the perturbation theorems follow Dirichlet-form / Rayleigh–Ritz and the matrix-tree theorem.

`zs_a24_v2_1_audit.py` (exact symbolic + Monte-Carlo, runnable; **54/54 computational, regression, and consistency checks PASS** — not 53 separate theorem proofs) covers, by Part: **Part I** — the general-graph stationary law, modular detailed balance, the exact H-theorem, the Schur-uniqueness inputs, the exact general spectrum A24.I-4G against the 121×121 build and the (2, 3, 4) control, and the two perturbation theorems (reciprocal Dirichlet sandwich, directional exact stationary/decay-rates/TV/forward-only gap,

and the full 121×121 microscopic Monte Carlo); **Part II** — the Gate-1 inner-action retraction, the Gate-3 partition $(6, 32, 83)/121$, the operator-dimension weights $(9, 4, 36)$ with $\Sigma = 49$, the F-A24.4 Bogomolnyi coefficient and global-vortex finding, the corpus-instanton ledger and per-edge retraction, and the trace/state split (corner trace $\pi = (3, 2, 6)/11$ vs $\omega = \tau(h \cdot) = (9, 4, 36)/49$); **Part III** — the no-go, the full quantum detailed balance of L_ω on M_{11} (KMS- and GNS-symmetric, modular-covariant, ρ_ω -stationary, residuals $\sim 10^{-18}$), and the microscopic interpolation family $L_{\{(s)\}}$ (stationarity, KMS pairing, and endpoints at $s = 0, 1/4, 1/2$); plus the anti-numerology face-triple test and cross-version safety. NOTE: "these are computational/regression/consistency verifications, **not** theorem proofs (the theorems carry their own proofs in the text); the unification is **finite-register**, and the **continuous-core dynamical lift is the single honest OPEN item** (F-A24.9); the surviving residuals — the Gate-2(ii) CLPW reference, the per-edge leg-count, ZS-F0's explicit state-sum, and B3 — are upstream, registered, or non-load-bearing." Independent rerun recommended. This work used AI tools (Anthropic Claude) for verification and drafting; the author assumes full responsibility.

Appendix E — Deep-Exploration Records (v1.2–v2.0, condensed)

Each version was developed by the standard exploration protocol (Step 0 long-list $\rightarrow$ Step 1 MECE issue-list $\rightarrow$ Step 2 issue-tree $\rightarrow$ Step 3 epistemic-status assignment $\rightarrow$ Step 4 convergence $\rightarrow$ Step 5 value). The records below are the condensed audit trail; status tags use the legend of §0.

v1.2 — the two §20 programs (Gate-3 partition; per-edge vortex). *Issue-list:* (I1) the $\mathbb{C}^{121}$ face partition; (I2) the per-edge instanton as a Bogomolnyi vortex. *Status:* I1 **DERIVED** — the coarse traces $(6, 32, 83)/121$ are (F(cube), F(truncated icosahedron), remainder) from ZS-F2 §11.4; I2 **CLOSED-NEGATIVE** — the minimal BPS vortex action is 2π , not $1/\kappa^2$, so the per-edge-vortex route fails. *Convergent;* value: grounded the Gate-3 partition, excluded a wrong instanton route.

v1.3 — closing the two §20 residuals. *Issue-list:* (I1) the per-edge action value; (I2) the observer-weight status; (I3) the $e^{\{-2/\kappa^2\}} \leftrightarrow \Lambda$ motivation. *Status:* I1 **RETRACTED** $\rightarrow$ **DERIVED** — $1/\kappa^2$ has no corpus instanton (the corpus actions are $\pi/A, 2\pi/A, 5\pi/A, 35\pi/3$); the corrected per-edge action is the $\mathbb{Z}_2$ seam-flip $t_{\text{flip}} = \pi/A$ (ZS-M3 §6), $S_{\{X \rightarrow Y\}} = 2\pi/A$; I2 **PROVEN** — $\omega = (9, 4, 36)/49 = d_{-i}^2/\Sigma d_{+j}^2$ two ways (ZS-A23.MC); I3 **RETRACTED** as numerology. *Convergent;* value: removed two over-claims, fixed the per-edge scale.

v1.4 — last internal closure + residual audit. *Issue-list:* (I1) the corner-trace identification; (I2) the per-edge same-seam lemma; (I3–4) the residual audit and C5. *Status (as recorded then):* I1 "DERIVED via Takesaki" — **later corrected in v1.5** (the dual weight is the trace times the modular density, so that argument fixes the *state* ω , not the trace); I2 **DERIVED** — the A24 seam is the ZS-F23/ZS-F0 §8.4 J-grading = the ZS-M3 $\mathbb{Z}_2$, fixing the scale at π/A ; residuals located as upstream/registered. *Convergent;* value: clarified residuals; its Gate-2 statement was the error v1.5 fixed.

v1.5 — finite-register trace/state unification + Gate-2 correction. *Issue-list:* (I1) the trace/state correction; (I2) the general exact spectrum; (I3) the two perturbation theorems. *Status:* I1 **DERIVED** — corner trace = matrix trace = $(3, 2, 6)/11 = \pi$; $\omega = (9, 4, 36)/49 = \tau(h \cdot)$ is the modular *state* (correcting the v1.4 conflation, restoring consistency with ZS-A23); no-go **PROVEN**; L_ω **PROVEN**; interpolation **DERIVED**; I2 **PROVEN** (Theorem A24.I-4G); I3 **PROVEN** (reciprocal + directional). *Convergent;*

value: fixed the central conflation, built the modular bridge — but over-stated it as "complete Part I $\leftrightarrow$ II unification."

v1.6/v2.0 — scope correction, full quantum detailed balance, and the v2.0 patch. *Issue-list:* (I1) bound the claim to finite-register; (I2) the full quantum DB theorem; (I3) symbol/textual fixes; (I4) audit strengthening. *Status:* I1 **DERIVED (finite) / OPEN (lift)** — PART III unifies π and ω on M_{11} and supplies the candidate dynamics; the continuous-core dynamical lift (a generator $\mathcal{L}_s$ on M_{obs} with $E \circ \mathcal{L}_s = \mathbf{L}_s \circ E$, modular covariance, CP) is **OPEN** (F-A24.9); I2 **PROVEN** (Theorem A24.III-1b: $\mathbf{L}_\omega$ is KMS- and GNS-symmetric on M_{11} , modular-covariant). The v2.0 patch additionally made the interpolation family a full microscopic QMS (Theorem A24.III-2), corrected the Schur uniqueness to "one scalar per edge + global edge-homogeneity," made the §7.3 full-121 $\times$ 121 microscopic Monte Carlo explicit, split the simulator into two protocols, fixed the trace/dual-weight and $\mathcal{L}_s/L_s$ expressions, and added the Fagnola–Umanità and quantum-Doob references. *Convergent;* value: A24 bounded to what is proved, with the single honest OPEN registered.

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Version History

Table 4. Version history.

Version	Date	Change
v1.0	June 2026	Initial draft. Four-gate continuous-clock lift. Contained a foundational error: claimed $M_{11} \rtimes_{\{\sigma^\omega\}} \mathbb{R}$ is Type II. Superseded.
v1.1	June 2026	Foundational revision. Retracted v1.0's finite-route Type-II error (inner modular flow $\Rightarrow$ Type I, CLOSED-NEGATIVE); rebuilt Part II on the genuine $III_1 \rightarrow II_\infty$ core with a II_1 corner and register embedding; absorbed the self-contained Part I (general theorem, H-theorem, full 121×121 spectrum, perturbation); redefined the Gate-3 target in continuous trace dimension. Audit <code>zs_a24_v1_1_audit.py</code> , 18/18 PASS . ($A, Q, \dim Z$) = (35/437, 11, 2) LOCKED .
v1.2	June 2026	§20 programs (Appendix E). Derived the Gate-3 partition (6, 32, 83)/121 from ZS-F2 §11.4; closed the per-edge Z-Anchor-vortex route as CLOSED-NEGATIVE (minimal BPS action $2\pi \neq 1/\kappa^2$); grounded the observer weights as operator dimensions d_i^2 . Audit 24/24 PASS. LOCKED .
v1.3	June 2026	Residual closure (Appendix E). Proved $\omega = (9, 4, 36)/49$ (ZS-A23.MC, two routes); resolved the equal-weight as combinatorial holography (ZS-F0 §8.5); retracted per-edge $= 1/\kappa^2$ (no corpus instanton) for the seam-flip π/A (ZS-M3 §6); retracted the $e^{\{-2/\kappa^2\}} \leftrightarrow \Lambda$ numerology. Audit 30/30 PASS. LOCKED .

Versio n	Date	Change
v1.4	June 2026	Internal closure + residual audit (Appendix E). Recorded a Takesaki argument for the corner trace (later corrected in v1.5 — it fixes the <i>state</i> , not the trace) and the same-seam lemma fixing the per-edge scale at π/A ; located the remaining items as upstream/registered. Audit 35/35 PASS. LOCKED.
v1.5	June 2026	Trace/state correction + finite-register bridge (Appendix E). Corrected the v1.0-era conflation: corner <i>trace</i> = matrix trace = $(3, 2, 6)/11 = \pi$, while $\omega = (9, 4, 36)/49 =$ the <i>state</i> $\tau(h \cdot)$ (restoring ZS-A23 consistency). New PART III : no-go, observer-balanced L_ω , and the interpolation joining L_π to L_ω . Upgraded §6 to the exact general spectrum A24.I-4G and §7 to two PROVEN perturbation theorems. Audit <i>zs_a24_v1_5_audit.py</i> , 42/42 PASS. LOCKED.
v1.6	June 2026	Scope correction + full quantum DB (Appendix E). Bounded the claim to finite-register unification (the continuous-core dynamical lift registered OPEN , F-A24.9); proved the full quantum detailed-balance theorem (A24.III-1b: L_ω KMS- and GNS-symmetric on M_{11} , modular-covariant); separated the e_a/q_a symbols; fixed the embedding-arrow, §11, §19, falsification-numbering, and amplitude/rate items; positioned the standard machinery honestly. Audit <i>zs_a24_v1_6_audit.py</i> , 48/48 PASS. LOCKED.
v2.0	June 2026	Consolidated public release (Appendix E). Editorial/mathematical patch on v1.6, no new physics: made the interpolation a full microscopic QMS family $L_{\{s\}}$ (Theorem A24.III-2, jumps $\sqrt{\gamma(h_j/h_i)^{s/2}}$, ρ_s , endpoints L_π, L_ω); corrected the GKLS uniqueness to one scalar per edge (Schur) + global edge-homogeneity ; made the full 121×121 microscopic Monte Carlo explicit (§7.3); split the simulator into two protocols (π via L_π , ω via L_ω); fixed the trace/dual-weight and $\mathcal{L}_s/L_s$ expressions; added the Fagnola–Umanità and quantum-Doob references; moved the deep-exploration records to Appendix E and compressed the body. Audit <i>zs_a24_v2_0_audit.py</i> , 53/53 PASS. (A, Q, dim Z) = (35/437, 11, 2) LOCKED. (Consolidated from internal Z-Spin Collaboration notes and external review v1.0 → v2.0.)
v2.1	June 2026	Editorial/mathematical patch of v2.0 (public release). No new physics; closes the final review items. (1) Γ_0 time-normalisation convention stated in §2 ($\Gamma_0 \equiv 1$ is a unit, not a fitted parameter — closing the last zero-free-parameter gap). (2) Subtitle added making explicit that the unification is finite-register and the continuous-core dynamical lift is left open (F-A24.9). (3) §10 embedding arrow corrected to $M_{11} \hookrightarrow M_{\text{obs}} \subset C_\omega$. (4) §U6 intertwining corrected to $E \circ \mathcal{L}_s = L_s \circ E$. (5) §11 status corrected — the conditional expectation E exists, but the dynamical intertwining is a requirement on the unconstructed lift and remains OPEN . (6) §7.3 figure stated as $\max_i \Delta p_i < 0.02$ (the audit PASS threshold; fixed-seed run ≈ 0.011) and described as a robustness illustration, not a certified tolerance. (7) §5 status downgraded to DERIVED (the per-edge Schur step is PROVEN; full uniqueness rests on the action-level global-edge-homogeneity input). (8) Audit strengthened (V7.6): full GNS- and KMS-symmetry of L_s at the intermediate $s = 1/4$ and all 36 jump-pair modular pairings, matching the §U4 text. (9) Residual v1.x/two-Part self-references replaced by v2.0/three-Part framing (Title/§1/§17/§20/Acknowledgements); the historical attributions in §12/§19 and the version trail are retained. Audit <i>zs_a24_v2_1_audit.py</i> , 54/54 PASS. (A, Q, dim Z) = (35/437, 11, 2) LOCKED.

— End of ZS-A24 v2.1 —